

## **MOTOR Guide to Estimating (Merged)**

Source start URL: <https://help.cccis.com/webhelp/motor/gte/guide.htm>

Captured pages: 148

### **1. ADAS Precautions**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/ADAS\\_PRECAUTIONS .htm](https://help.cccis.com/webhelp/motor/gte/Content/ADAS_PRECAUTIONS.htm)

#### **ADAS Precautions**

On vehicles that use advanced driver-assistance systems (ADAS), OEM repair information must be consulted to ensure repairs will not negatively impact ADAS functionality.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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### **2. Add If Required**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/ADD IF REQUIRED.htm>

#### **Add If Required**

MOTOR Collision Estimating Data is based on the base model vehicle configuration, standard or regular production options, and/or standard replacement operations. "Add if required" operations are for extra procedures necessitated by optional factory equipment or certain collision scenarios that may be encountered. "Add if required" operations should be added to the estimate whenever applicable after an "on the spot" inspection of vehicle damage and/or vehicle options.

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### **3. Air Conditioner Refrigerant**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/AIRCONDITIONERREFRIGERANT.htm>

#### **Air Conditioner Refrigerant**

Reported refrigerant capacities are FOR ESTIMATING PURPOSES ONLY. ALWAYS refer to the vehicle AC label or manufacturer's service manual for recommended repair procedures and system capacity.

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## **4. Air Conditioning System – Recharge R1234YF**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/AIRCONDITIONINGSYSTEM\\_RECHARGE\\_R1234YF.htm](https://help.cccis.com/webhelp/motor/gte/Content/AIRCONDITIONINGSYSTEM_RECHARGE_R1234YF.htm)

### **Air Conditioning System – Recharge R1234YF**

Attach/detach equipment

Checking for leaks using electronic leak detector

Equipment operation

Vacuum leak test

Adding required oil manually

Equipment maintenance

Cost of refrigerant & oil

Cost of seals or O-rings

Leak diagnosis

Refrigerant recovery

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## **5. Air Conditioning System – Recharge R134A**

Source:

[https://help.cccis.com/webhelp/motor/gte/Content/AIRCONDITIONINGSYSTEM\\_RECHARGE\\_R134A.htm](https://help.cccis.com/webhelp/motor/gte/Content/AIRCONDITIONINGSYSTEM_RECHARGE_R134A.htm)

### **Air Conditioning System – Recharge R134A**

Attach/detach equipment

Equipment operation

Adding required oil  
Cost of refrigerant & oil  
Cost of seals or O-rings  
Equipment maintenance  
Leak diagnosis  
Refrigerant recovery  
Test for leaks  
Retrofit for R134A

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## **6. Acoustical and Structural Foam Fillers**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/Acoustical\\_And\\_Structural\\_Foam\\_Fillers.htm](https://help.cccis.com/webhelp/motor/gte/Content/Acoustical_And_Structural_Foam_Fillers.htm)

### **Acoustical and Structural Foam Fillers**

This type of product is used by some OEMs to control NVH (noise vibration and harshness) or add structural integrity to the vehicle body. Acoustical foam is a soft to firm, pliable product while structural foam is a hard, dense, rigid product. Acoustical and/or structural fillers may be found internally on vehicle panels; therefore, identification of foam type and location may require an inspection after vehicle disassembly. Foam products must be removed if within 6 inches of an open flame, excessive heat (600 degrees) or welding operation. Any foam damaged or removed during the repair process must be replaced. Always use the correct replacement product.

#### **SPECIAL NOTATION:**

Not all vehicles contain foam fillers; others may contain foam fillers in one or more locations. See figures below for general foam fillers location. Refer to OEM for specific repair recommendations / procedures and replacement product(s).

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## **7. Adhesive Type Moldings, Nameplates & Emblems**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/AdhesiveTypeMoldings.htm>

### **Adhesive Type Moldings, Nameplates & Emblems**

Clean/heat surfaces

Component align/position

Installation of component only

Adhesive backing removal,

Clean up and replace

Damaged panel repair

Door hardware

Refinishing prior to application

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## **8. Air Conditioner**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/AirConditioner.htm>

### **Air Conditioner**

#### **SPECIAL NOTATION:**

A "Refrigerant Recovery" procedure should be considered prior to any repair necessitating a refrigerant line disconnect from an AC system containing refrigerant. Published recovery time is for a service technician to operate and periodically monitor typical refrigerant recovery & recycling equipment as the vehicle's existing refrigerant is transferred out of the vehicle and into the machine (0.3) and reprocessed by the machine (0.1).

Recovery time does not include overall recycling machine processing time, nor does it include "AC Service" which is the evacuation, recharge and testing of an AC system following repairs.

R&R time is listed with parts

Cost of refrigerant & oil

Cost of seals or O-rings

Refrigerant recovery

Test for leaks

Retrofit for 134a

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## 9. Back Glass

Source: <https://help.cccis.com/webhelp/motor/gte/Content/BACKGLASS.htm>

### Back Glass

#### SPECIAL NOTATION:

Normal service procedures on some applications. MOTOR advises that these factors be considered before performing glass operations in the event unavoidable damage occurs. Time for the removal of old urethane, clean and preparation of sealing surfaces (vehicle and/or glass) should be estimated after an on-the-spot evaluation.

Reveal & interior moldings / trim

Rubber channel (if installed)

Test for leaks

Urethane adhesive application

Wiper arm

Clean up of broken glass

Cost of installation material or kit

High-mounted stop lamp

Refinish glass pinchweld (epoxy primer)

Things to Consider

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

NAGS - Glass Prices

R & I (Remove & Reinstall)

Structural Glass

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## 10. Bolted On Parts

Source: [https://help.cccis.com/webhelp/motor/gte/Content/BOLTED\\_ON\\_PARTS.htm](https://help.cccis.com/webhelp/motor/gte/Content/BOLTED_ON_PARTS.htm)

### Bolted On Parts

The phrase "BOLTED ON PARTS" referenced in this publication refers to mechanically fastened components that are directly attached to the component that is being replaced and includes braces, brackets, mounts, shields, extensions, interior trim, weatherstrips etc. Mechanical fasteners may include bolts, nuts, screws, clips and rivets.

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## **11. Bagging (Cover Entire Vehicle Exterior)**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/Bagging\\_CoverEntireVehicleExterior\\_.htm](https://help.cccis.com/webhelp/motor/gte/Content/Bagging_CoverEntireVehicleExterior_.htm)

### **Bagging (Cover Entire Vehicle Exterior)**

Published refinish times include time necessary to mask exterior surface adjacent the refinish area to a perimeter of 36 inches, or 3 feet. When the process of perimeter masking is substituted for an entire vehicle bagging procedure, then no additional time should be added. If entire vehicle bagging is used along with perimeter masking, then the following formula may be considered.

#### **APPLY AND REMOVE VEHICLE COVER (BAGGING)**

- Add 0.2 each time a cover is applied and removed

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## **12. Basic Color Coat Application**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/BasicColorCoatApplication.htm>

### **Basic Color Coat Application**

Back tape opening (handle, lock cylinder, mirror)

Clean component (solvent wash)

Clean sprayer

Color coat application

Initial dry sand (as recommended by paint manufacturer)

Light buff, lacquer paint only

Load sprayer

Mask adjacent panels (3 foot perimeter)

Mask / close gap between adjacent panels up to foam tape (over spray)

Mask glass opening

Mask / protect grille radiator opening (over spray)

Mix pre-made color with necessary solvents

Primer-Sealer coat application

Primer-Sealer coat final clean

Primer-Sealer coat final application

Remove masking

Retrieve accurate color information, including paint chip

Adhesion promoter (unprimed flexible component)

Backside refinishing

Blending into adjacent panels

Color matching to adjacent panels

Cover mask engine / compartment to prevent overspray

Cover / mask entire exterior of vehicle to prevent overspray damage

Cover / mask for prime and block

Cover / mask for cut-in

Cover / mask recessed edges / jambs

Cover / mask trunk / compartment to prevent overspray

Cover / mask interior of vehicle to prevent overspray damage

Edge refinishing

Flash/wait time between material applications

Grind, fill & smooth welded seams

(up to 150 grit sandpaper)

Paint or material costs

Prime & block (high build / primer-filler)

Test spray-out panel

Tinting Primer-Sealer

Tinting to achieve color match

Underside refinishing

Weld, grind or sanding damage to adjacent panels

Wet sanding

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### **13. Bleed Brake Hydraulic System**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/BleedBrakeHydraulicSystem.htm>

#### **Bleed Brake Hydraulic System**

SPECIAL NOTATION:

Bleed Brake System operation times are based upon other suspension and/or brake work already being performed and the Bleed Brake System Operation performed in conjunction with those other operations.

- Bleed brake system
- Add fluid to master cylinder reservoir
- Brakes adjust
- Cost of brake fluid

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### **14. Bolt Or Clip-On Moldings**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/BoltOrClip\\_OnMoldings.htm](https://help.cccis.com/webhelp/motor/gte/Content/BoltOrClip_OnMoldings.htm)

#### **Bolt Or Clip-On Moldings**

CONSIDERED:

- Drill time, if required
- If new parts are used on a new panel,  
add one half the replacement time
- If old parts are used on a new panel,  
add entire replacement time

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## 15. Center Pillar

Source: <https://help.cccis.com/webhelp/motor/gte/Content/CENTERPILLAR.htm>

### Center Pillar

#### SPECIAL NOTATION:

When replacing welded rear door hinges, add one half of the Hinge R&R time to Center Pillar R&R time.

Adhesive application if required (all types)

Carpet & insulation turn back

Caulk/seam sealer

Cutting & welding as necessary

Door striker

Grind, fill & smooth welded seams (up to 150 grit sandpaper)

Interior trim panels

Rivet removal and installation if required

Seat belt anchor

Sill plates

Weatherstrips

Airbag components

Carpet, insulation or seat R&I

Doors

Door hinge

Electrical wiring

Emblems & nameplates

Foam

Headliner

Hinge plate transfer

Lamp

Moldings

Roof cover

Stripe tape, decals, or overlay

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## 16. Clear Coat Undamaged Panel

Source: [https://help.cccis.com/webhelp/motor/gte/Content/CLEAR\\_COAT\\_UNDAMAGED\\_PANEL.htm](https://help.cccis.com/webhelp/motor/gte/Content/CLEAR_COAT_UNDAMAGED_PANEL.htm)

### Clear Coat Undamaged Panel

#### SPECIAL NOTATION:

Calculations for clear coating an undamaged panel are based upon the outer surface only and should not include additions for underside, inside, or edges of the clear-coated panel. There should be no overlap deduction between refinished or clear-coated panel(s), nor should this procedure be applied towards the maximum clear coat allocation. Clear coating may be necessary for adjacent body panel(s) to nearest break point (see Refinish Clear Coat Application ). The following formula may be considered in the event this type of procedure is required on an undamaged panel.

- Each clear coated panel(s) 40% of panels Base Refinish Time

Back tape opening (handle, lock cylinder, mirror)

Bonding / adhesion coat application (if required)

Clear coat application

Clean component (solvent / detergent wash)

Clean in preparation for material application

Initial wet sand or scuff

Mask adjacent panels (3 foot perimeter)

Mask / close gap between adjacent panels up to foam tape (over spray)

Mask glass opening

Mask / protect grille radiator opening (over spray)

Remove masking

Tack wipe surface (when required)

Correction of pre-existent surface imperfections

Damage repair

De-nib / wet sand and/or buff for polishing

Flash/wait time between material applications

Masking of attached parts

Material costs

R&I of attached parts

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## 17. Console - Floor or Overhead

Source: [https://help.cccis.com/webhelp/motor/gte/Content/CONSOLE\\_FLOOROROVERHEAD.htm](https://help.cccis.com/webhelp/motor/gte/Content/CONSOLE_FLOOROROVERHEAD.htm)

## **Console - Floor or Overhead**

Console

Emblems & nameplates

Gauges

Moldings & ornamentation

Shift lever (if necessary)

Front seat R&I

Headliner R&I

Instrument panel

Straightening inner panels, brackets or braces

Sunvisors

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## **18. Radiator**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/COOLING\\_RADIATOR.htm](https://help.cccis.com/webhelp/motor/gte/Content/COOLING_RADIATOR.htm)

### **Radiator**

#### **SPECIAL NOTATION:**

If R&R of a coolant hose, connected to both the radiator and water pump, is combined with removal of the radiator and water pump, then no additional time should be estimated for R&R or R&I of that coolant hose.

Drain, and refill fluids

Check fluids using a dipstick

Electric fan assembly R&I

Purge air from system (one time)

Belts

Check fluids using scan tool

Cost of coolant

Electric fan assembly O/H or R&R

Mechanical fan

Pulley

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## **19. Radiator Support**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/COOLING\\_RADIATORSUPPORT.htm](https://help.cccis.com/webhelp/motor/gte/Content/COOLING_RADIATORSUPPORT.htm)

### **Radiator Support**

Special Notation:

Operations / components included with R&R radiator support are listed in FOOTNOTES in each chapter and below as INCLUDED.

Caulk / seam sealer

Included parts are listed in footnote in each chapter

Welding as necessary

See " ESTIMATED WORK TIME PREMISE" DOES NOT INCLUDE

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## **20. Camera and Sensor Aiming**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/CameraSensorAiming.htm>

### **Camera and Sensor Aiming**

Connecting scan tool

Measure for and setting up targets

Physical alignment of Camera/Sensor

Putting the vehicle in position - dedicated area or alignment rack

Use of scan tool for aiming procedure

Adjusting tire pressure

Checking levelness of the ground

Checking tire runout

Creating targets /assembling targets

Filling the gas tank

Installing fixtures and positioning on wheels

R&I of parts for alignment

Removing/replacing customer valuables from vehicle

Repair of any mounting surfaces

Road test - dynamic aiming

Wheel alignment

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## **21. Clear Coat Finishes (Base Coat / Clear Coat)**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/ClearCoatFinishes\\_BaseCoat\\_ClearCoat\\_.htm](https://help.cccis.com/webhelp/motor/gte/Content/ClearCoatFinishes_BaseCoat_ClearCoat_.htm)

### **Clear Coat Finishes (Base Coat / Clear Coat)**

#### **SPECIAL NOTATION:**

The following items or operations were not considered during the development of any published basic refinish operation times. If any of these items or operations are required, they should be considered by the estimator. Calculations should be made after deductions for Overlap, and additions for underside and edges if required. It should be noted that it is not possible to sand, de-nib & polish, or rub-out and buff a matte clear finish, without changing the finish from matte to gloss. Therefore, a refinishing booth without airborne contaminants is required for this operation.

- First major panel;
  - Add 40% to refinish time
- Each additional panel;
  - Add 20% to refinish time
- Maximum time allocation
  - 2.5 hours

All components clear coated during a single, continuous procedure

Apply clear coat

Clean sprayer (1 time)

Mix clear coat (1 time)

Tack wipe surface (when required)

Any component clear coated as separate procedure

Any operation previously excluded in "Refinish time Premise" and/or "Basic Color Coat Applications" groups

Flash/wait time between material applications

Material costs

Texture matching (matte clear finishes)

Test spray-out panel

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## **22. Color Blend (Adjacent Panels)**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/ColorBlending\\_AdjacentPanels\\_.htm](https://help.cccis.com/webhelp/motor/gte/Content/ColorBlending_AdjacentPanels_.htm)

### **Color Blend (Adjacent Panels)**

Estimated refinish times for color blending should defer to the judgment of an estimator or appraiser following an on-the-spot evaluation of the specific vehicle and refinish requirements in question.

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## **23. Component Classification**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/ComponentClassification.htm>

### **Component Classification**

The purpose of classifying components is to describe physical properties of the component. Indicators are not intended to determine labor rates charged, or to be inclusive of all components. If using component classifications to determine technician compensation, this would be at the sole discretion of the repair facility.

CLASSIFICATION INDICATORS ARE PROVIDED FOR YOUR CONVENIENCE AND MUST ONLY BE CONSIDERED A HELPFUL GUIDE.

MOTOR component classifications are defined as follows:

(M) Mechanical: Components that transform one form of motion or energy into another. Mechanical components would likely be serviced at a mechanical service facility rather than a body repair facility if that component failed during normal operation.

(S) Structural: Components that provide a load bearing foundation for the purpose of safety and/or stability.

(no classification) Body: Components that do not fall under the mechanical or structural classification.

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## **24. Cowl, Hinge Pillar & Dash Panel**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/Cowl\\_HingePillar.htm](https://help.cccis.com/webhelp/motor/gte/Content/Cowl_HingePillar.htm)

### **Cowl, Hinge Pillar & Dash Panel**

Adhesive application if required (all types)

Carpet & mats turn back

Caulk/seam sealer

Cowl side trim

Cutting & welding as necessary

Grind, fill & smooth welded seams (up to 150 grit sandpaper)

Hinge pillar glass and moldings

Jamb switch

Rivet removal and installation if required

Sill plates

Weatherstrips

Air bags

Air conditioning parts

Bolted on parts

Carpet, insulation & seat R&I

Door

Door hinge

Electrical wiring

Engine

Fender

Foam

Headliner

Hinge plate transfer

Instrument panel or crash pad

Interior moldings

Parking brake assembly

Roof rail molding

Windshield glass

Windshield reveal moldings

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## **25. Damaged or Non-Reusable Moldings, Components and Fasteners**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/DAMAGEDMOLDINGS.htm>

### **Damaged or Non-Reusable Moldings, Components and Fasteners**

Some moldings may be damaged or broken during normal R&I procedures. Additionally, some plastic or metal fasteners may be damaged and their ability to be reused jeopardized because of single-use design, age or exposure to the elements. Moldings, emblems and trim attached using a heat stake method are considered to be non-reusable. Caution should be taken while removing all fasteners, both metal and plastic. Some components are identified by the OE as single use/non-reusable. Those components are identified by a footnote applied to the main component(s) and identifies any attaching component(s) that may also be single use/non-reusable. MOTOR recommends these factors be considered and mutually agreed upon before finalizing any repair cost estimate.

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## **26. De-Nib & Polish**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/DE\\_NIB\\_POLISH.htm](https://help.cccis.com/webhelp/motor/gte/Content/DE_NIB_POLISH.htm)

### **De-Nib & Polish**

#### **SPECIAL NOTATION:**

Refinished panels may or may not require a varying amount of de-nibbing, a process used to remove small particles in final finish surface. The clear coat contains ultraviolet screeners and reducing the clear coat thickness (mils) may result in early paint failure. Follow vehicle manufacturer's recommendations when performing this type of repairs. Calculations should be based upon the base refinish time outer surface only and should not include additions for clear coat, underside, inside or edges. In the event that this type of operation will be performed, MOTOR suggests the following formula be considered:

Each panel requiring de-nibbing (refinish or blend)

HOOD, ROOF, TRUNK LID, SPOILER

First panel add up to 20% of full base refinish time, each additional panel add up to 10%

FENDER, DOOR, QUARTER PANEL, BUMPER COVER

First panel add up to 10% of full base refinish time, each additional panel add up to 5%

Panel outer surface only

Paint nib removal as required (spot only)

Spot polish only

Acid rain damage

Full panel polish

Overspray removal

Removal of residual material from recessed edges and jambs if required

Scratch damage

Wash, clean, wax or detail entire vehicle prior to delivery if required

Wet sand full panel

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## **27. Dis/Reasmble to Refn**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/DISREASSMBLETOREFN.htm>

### **Dis/Reasmble to Refn**

The disassemble and reassemble operation present for certain assemblies reflects a condition where the OEM offers a new replacement assembly that must be disassembled and reassembled for refinish purposes.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **28. Door - Door Shell R&R Sliding Type**

Source:

[https://help.cccis.com/webhelp/motor/gte/Content/DOOR\\_DOOR\\_SHELL\\_R\\_R\\_SLIDING\\_TYPE.htm](https://help.cccis.com/webhelp/motor/gte/Content/DOOR_DOOR_SHELL_R_R_SLIDING_TYPE.htm)

### **Door - Door Shell R&R Sliding Type**

## SPECIAL NOTATION:

Some replacement components may or may not be supplied with duplicated OEM caulk / seam sealer. This is not included in R&R time and requires an on-the-spot evaluation. Disconnect at hinges (unless otherwise noted in text)

Access cover

Align to vehicle

Applique

Belt molding

Bonded glass installation

Glass (fixed / moveable)

Glass reveal moldings

Inside handles and bezels

Internal mechanisms

Latch mechanisms

Lock and cylinder assembly (if necessary)

Outside handle

Speakers

Trim panel R&I

Vents

Watershield

Weatherstrips (if attached to door)

Body side moldings

Bonded glass removal from damaged door shell

Emblems and nameplates

Lock cylinder recoding

Moldings

Removal of shipping brace(s)

Stripe tape, decals or overlays.

Sunshade

Things to Consider

Add If Required

Adhesive Type Moldings, Nameplates & Emblems

Damaged or Non-Reusable Moldings and Fasteners

Decals, Stripe Tape & Overlays

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise  
R & I (Remove & Reinstall)  
Special Substrate Metals  
Refinish Starting Point  
Basic Color Coat Application  
Parts Supplied In Color  
Refinish Clear Coat Application  
Refinish Time Listings  
Refinish Time Premise  
Refinish Processes  
Clear Coat Finishes (Base Coat / Clear Coat)  
Color Blend (Adjacent Panels)  
Deductions to Basic Refinish Times (Refinish Overlap)  
Panel and/or Component Designation  
Two-Tone Refinishing (Second Color Tone Application)  
Things to Know  
Bagging (Cover Entire Vehicle Exterior)  
De-Nib & Polish  
Edges Of New Parts (Edging)  
Stone Chip Guard (Protective Material)  
Three-Stage Color Blend (Adjacent Panels)  
Three-Stage Finishes (Base / Mica / Clear Coat)  
Wet / Dry Sand, Rub-Out & Buff

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **29. Door - Hinged and Sliding Type Outer Panel R&R**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/DOOR\\_OUTERPANELR\\_R.htm](https://help.cccis.com/webhelp/motor/gte/Content/DOOR_OUTERPANELR_R.htm)

### **Door - Hinged and Sliding Type Outer Panel R&R**

#### **SPECIAL NOTATION:**

Some cutting or welding processes may require the door glass to be removed prior to replacing outer panel. This factor should be considered prior to finalizing any repair cost estimates. Disconnect wiring at connector if accessible.

Adhesive application if required (all types)

Align to vehicle

Applique

Belt molding

Caulk / seam sealer

Cutting & welding as necessary

Door R&I

Flutter material application if required

Grind, fill and smooth welded

seams (up to 150 grit sandpaper)

Inside handles & bezels

Lock and cylinder assembly (if necessary)

Mirror

Outside door handle

Rivet removal and installation if required

Speakers

Trim panel R&I

Water shield

Weatherstrips (if attached to door)

Back glass wiper or washer

Body side moldings

Clean up broken glass

Electrical wiring

Emblems & nameplates

Glass

Hinge

Lock cylinder recoding

Moldings

Passive restraint system

Power lock option

Power window option

Regulator

Reinforcement bar

Straightening of door frame

Stripe tape, decals or overlays

Sunshade

Things to Consider

Add If Required  
Adhesive Type Moldings, Nameplates & Emblems  
Damaged or Non-Reusable Moldings and Fasteners  
Decals, Stripe Tape & Overlays  
Footnotes  
Included and/or Not Included Labor Operations  
Labor Time Listings  
Labor Time Premise  
R & I (Remove & Reinstall)  
Special Substrate Metals  
Refinish Starting Point  
Basic Color Coat Application  
Parts Supplied In Color  
Refinish Clear Coat Application  
Refinish Time Listings  
Refinish Time Premise  
Things to Know  
Bagging (Cover Entire Vehicle Exterior)  
Clear Coat Undamaged Panel  
De-Nib & Polish  
Edges Of New Parts (Edging)  
Stone Chip Guard (Protective Material)  
Three-Stage Color Blend (Adjacent Panels)  
Three-Stage Finishes (Base / Mica / Clear Coat)  
Wet / Dry Sand, Rub-Out & Buff

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **30. D & R (Disconnect and Reconnect)**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/D\\_R\\_Disconnect\\_Reconnect\\_.htm](https://help.cccis.com/webhelp/motor/gte/Content/D_R_Disconnect_Reconnect_.htm)

### **D & R (Disconnect and Reconnect)**

Some labor procedures require disconnect (unplug and/or unbolt) of a component / assembly at the point where it is attached to the subject part; the component assembly is not completely removed from the vehicle. The component is reconnected during the assembly procedure. Due to various configurations and type of parts that may be involved in the D & R operations, and considering that the times involved are generally not definable in tenths of hours, time for D & R is not provided. When D & R is necessary to perform a labor operation, it is included in the estimated work time.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **31. Decals, Stripe Tape & Overlays**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/Decals\\_StripeTape\\_Overlays.htm](https://help.cccis.com/webhelp/motor/gte/Content/Decals_StripeTape_Overlays.htm)

### **Decals, Stripe Tape & Overlays**

Clean/heat surfaces

Component align/position

Installation of component only

Damage panel repair

Door hardware

Emblems & nameplates

Moldings & ornamentation

Mounting adhesive removal

Refinishing prior to application

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **32. Deductions to Basic Refinish Times (Refinish Overlap)**

Source:

[https://help.cccis.com/webhelp/motor/gte/Content/DeductionsToBasicRefinishTimes\\_RefinishOverlap\\_.htm](https://help.cccis.com/webhelp/motor/gte/Content/DeductionsToBasicRefinishTimes_RefinishOverlap_.htm)

### **Deductions to Basic Refinish Times (Refinish Overlap)**

OVERLAP - NON-ADJACENT PARTS:

- First major panel;
- Use full published time
- Each additional part;
- Deduct 0.2 per part

#### OVERLAP - ADJACENT PARTS:

- First major panel;
- Use full published time
- Each additional part with a base time of 1.0 hour or greater;
- Deduct 0.4 per part
- Each additional part with a base time less than 1.0 hour;
- Deduct 0.2 per part

#### OVERLAP - INNER PANEL COMPONENTS:

- First inner part;
- Use full published time
- Each additional inner part with a base time of 0.5 hour or greater;
- Deduct 0.2 per part
- Inner part with a base time less than 0.5 hour;
- No deduct

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **33. Door - Door Shell R&R, Hinged Type**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/Door\\_DoorShellR\\_R.htm](https://help.cccis.com/webhelp/motor/gte/Content/Door_DoorShellR_R.htm)

### **Door - Door Shell R&R, Hinged Type**

#### SPECIAL NOTATION:

Some replacement components may or may not be supplied with duplicated OEM caulk / seam sealer, this is not included in R&R time and requires an on-the-spot evaluation.

Disconnect at hinges (unless otherwise noted in text).

Disconnect wiring at connector if accessible.

Align to vehicle

Applique

Belt Molding

Bonded glass installation  
Door check  
Electrical wiring (standard features)  
Glass (fixed / moveable)  
Hinge halves bolted to door  
Inside handles & bezels  
Internal mechanisms  
Latch mechanisms  
License lamp (if attached to door)  
License plate (if attached to door)  
Lock & cylinder assembly (if necessary)  
Mirror  
Outside handle  
R&I Door Module (if applicable)  
Regulator  
Speakers  
Trim panel R&I  
Vents  
Water shield  
Weatherstrips (if attached to door)  
Window \ run channel  
Back glass wiper or washer  
Body side moldings  
Bonded glass removal from damaged door shell  
Electrical wiring (if not detachable from main body harness)  
Emblems & nameplates  
Hinge halves bolted to pillar  
Hinge (welded type)  
Lock cylinder recoding  
Moldings  
Passive restraint system  
Power lock option  
Power window option  
Removal of shipping brace(s)  
Spare tire carrier  
Stripe tape, decals or overlays  
Sunshade

Things to Consider

Add If Required

Adhesive Type Moldings, Nameplates & Emblems

Damaged or Non-Reusable Moldings and Fasteners

Decals, Stripe Tape & Overlays

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

R & I (Remove & Reinstall)

Special Substrate Metals

Refinish Starting Point

Basic Color Coat Application

Parts Supplied In Color

Refinish Clear Coat Application

Refinish Time Listings

Refinish Time Premise

Refinish Processes

Clear Coat Finishes (Base Coat / Clear Coat)

Color Blend (Adjacent Panels)

Deductions to Basic Refinish Times (Refinish Overlap)

Panel and/or Component Designation

Two-Tone Refinishing (Second Color Tone Application)

Things to Know

Bagging (Cover Entire Vehicle Exterior)

Clear Coat Undamaged Panel

De-Nib & Polish

Edges Of New Parts (Edging)

Stone Chip Guard (Protective Material)

Three-Stage Color Blend (Adjacent Panels)

Three-Stage Finishes (Base / Mica / Clear Coat)

Wet / Dry Sand, Rub-Out & Buff

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## 34. Drilling Operations

Source: <https://help.cccis.com/webhelp/motor/gte/Content/DrillingOperations.htm>

### Drilling Operations

Service components may be inconsistently delivered by the manufacturer with mounting holes drilled at the factory. Drilling time should be added to the estimate if a component is received without mounting holes pre-drilled.

Time to drill holes is listed in parentheses next to the part description, for example, (d.2) indicates 0.2 for drilling. This time should be added to the estimate when it is determined a drilling operation is required. Time shown is for round holes only. Estimate time accordingly if holes must be a shape other than round.

CONSIDERED:

- If new parts are used on a new panel, add drilling time and one half the replacement time
- If old parts are used on a new panel, add drilling time and entire replacement time

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## 35. Electronic Systems & On-Board Computers

Source:

[https://help.cccis.com/webhelp/motor/gte/Content/ELECTRONICSYSTEMS\\_ON\\_BOARDCOMPUTERS.htm](https://help.cccis.com/webhelp/motor/gte/Content/ELECTRONICSYSTEMS_ON_BOARDCOMPUTERS.htm)

### Electronic Systems & On-Board Computers

Special caution should be taken when servicing vehicles due to the wide use of computer modules, electronic sensors and printed circuit boards. This type of equipment is very sensitive to high temperatures and voltage fluctuations, including electrostatic discharge. A few safety tips you should consider are:

- Disconnect battery cable
- Computer modules should not be exposed to temperatures exceeding O.E.M. recommendations
- Remove component if necessary
- Discharge static electricity before handling component(s)
- Avoid touching electrical terminals of component

I-CAR's Uniform Procedures for Collision Repair states that electronic components should be removed from the vehicle if welding is to take place within 12 inches of the component. Refer to manufacturer's specific recommendations for each vehicle.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **36. Estimating Sequence**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/ESTIMATINGSEQUENCE.htm>

### **Estimating Sequence**

Listed under each basic assembly are the related and component parts. List the damage according to the basic assembly, taking the basic item first. Then, starting from the outside of the vehicle and working inward, list everything attached to it which is damaged. By following this procedure, you will make a more thorough inspection.

For instance, on an estimate involving front fender damage, inspect as shown:

Fender

molding

nameplate

emblem

Fender Liner

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **37. Edges Of New Parts (Edging)**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/EdgesOfNewParts\\_Edging\\_.htm](https://help.cccis.com/webhelp/motor/gte/Content/EdgesOfNewParts_Edging_.htm)

### **Edges Of New Parts (Edging)**

#### **SPECIAL NOTATION:**

The following items or operations were not considered during the development of any published basic refinish operation times. If any of these items or operations are required, they should be considered by the estimator. An additional paint mix is required if the edge color is a different color than the exterior body color. Clear coat (gloss or matte) will be required for base color coat applications.

Refer to specific parts text for estimated time allowance

Use full refinishing time without deduction for overlap

Clear coat

Color tinting

Mixing a different edge color

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **38. Engine / Transaxle Assembly R&I**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/Engine\\_TransaxleAssemblyR\\_I.htm](https://help.cccis.com/webhelp/motor/gte/Content/Engine_TransaxleAssemblyR_I.htm)

### **Engine / Transaxle Assembly R&I**

#### CONVENTIONAL FRAME VEHICLES

Air cleaner

Hoist or lift attach

Splash shields (if necessary)

A/C condenser

Drain & replace oil and fluids

Evacuate & recharge A/C system

Linkage adjustments

Radiator

Replacement of any parts

Road Test

#### UNITIZED FRAME VEHICLES

Air cleaner

R&I Engine/Transaxle with R&I engine cradle and suspension (both sides) as one unit

Hoist or lift table attach

Splash shields (if necessary)

A/C condenser

Drain & replace oil and fluids

Evacuate & recharge A/C system

Linkage adjustments

Radiator

Replacement of any parts

Road Test Vehicle

Road Wheel

Suspension/wheel alignment

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **39. Fluids**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/FLUIDS.htm>

### **Fluids**

Fluids listed are FOR ESTIMATING PURPOSES ONLY. ALWAYS refer to the OEM service information to verify fluid type, grade and capacity.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **40. Frame Crossmembers R&R - Front, Rear or Suspension Mount, Bolted Type**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/FRAMECROSSMEMBERS\\_FRONT\\_REARORSUSPENSIONMOUNT\\_BOLTEDTYPE.htm](https://help.cccis.com/webhelp/motor/gte/Content/FRAMECROSSMEMBERS_FRONT_REARORSUSPENSIONMOUNT_BOLTEDTYPE.htm)

### **Frame Crossmembers R&R - Front, Rear or Suspension Mount, Bolted Type**

Crossmember R&R

Attached parts R&I

Support engine /powertrain (if required )

Attached parts R&R

Brakes bleed or adjust

Suspension / wheel alignment

Exhaust system components

Fuel tank

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **41. Frame Crossmembers R&R - Front, Rear or Suspension Mount, Welded Type**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/FRAMECROSSMEMBERS\\_FRONT\\_REARORSUSPENSIONMOUNT\\_WELDEDTYPE.htm](https://help.cccis.com/webhelp/motor/gte/Content/FRAMECROSSMEMBERS_FRONT_REARORSUSPENSIONMOUNT_WELDEDTYPE.htm)

### **Frame Crossmembers R&R - Front, Rear or Suspension Mount, Welded Type**

Attached parts R&I

Support engine /powertrain (if required)

Welding as necessary

Attached parts R&R

Brakes adjust & bleed

Suspension / wheel alignment

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **42. Frame - Conventional Frame Vehicles**

Source:

[https://help.cccis.com/webhelp/motor/gte/Content/FRAME\\_CONVENTIONALFRAMEVEHICLES.htm](https://help.cccis.com/webhelp/motor/gte/Content/FRAME_CONVENTIONALFRAMEVEHICLES.htm)

### **Frame - Conventional Frame Vehicles**

Bed assembly R&I (truck)

Body assembly R&I

Brackets (bolted on)

Bumper assemblies Front / Rear R&I

Bushings

Clips

Drive shaft

Drivetrain as assembly R&I

Engine & transmission R&I

Front sheet metal R&I (as an assembly when practical)

Fuel tank (primary)

R & I independent rear suspension (LT & RT side)

assemblies, axles and housing)  
Stabilizer bar  
Steering gear/gear assembly R&R/R&I  
Suspension assemblies R&I  
Transfer lines to new assembly  
Transfer steering gear and linkage as an assembly  
Auxiliary fuel tank (second tank)  
Brackets (weld on)  
Brakes adjust & bleed  
Electrical wiring  
Exhaust systems  
Fluids (top off)  
Heat shields  
Interior trim / carpet  
Parking brake adjust  
Power steering cooler  
Pulling or straightening time  
Seats  
Shift & throttle linkage adjust  
Special equipment transfer  
Steering gear individual components R&R/R&I  
Suspension / wheel alignment  
Trailer hitch  
Transfer fuel (hold tank if necessary)  
Transmission cooler  
Wheels

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **43. Frame - Unitized Frame Vehicles**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/FRAME\\_UNITIZEDFRAMEVEHICLES.htm](https://help.cccis.com/webhelp/motor/gte/Content/FRAME_UNITIZEDFRAMEVEHICLES.htm)

### **Frame - Unitized Frame Vehicles**

Caulk / seam sealer  
Welding as necessary  
Adjacent panels  
Body sheet metal  
Bolted on parts (brackets & braces)  
Brakes adjust & bleed  
Drive train assembly  
Electrical wiring  
Floor mats, insulation, & trim  
Pulling or straightening time  
Set up on frame machine & diagnosis  
Suspension / wheel alignment

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **44. Front Suspension - Individual Operations**

Source:

[https://help.cccis.com/webhelp/motor/gte/Content/FRONTSUSPENSION\\_INDIVIDUALOPERATIONS.htm](https://help.cccis.com/webhelp/motor/gte/Content/FRONTSUSPENSION_INDIVIDUALOPERATIONS.htm)

### **Front Suspension - Individual Operations**

Raise & support vehicle (1 time)  
Brake calipers R&I  
Brake pads or shoes  
Brakes adjust & bleed  
Road wheel  
Suspension / wheel alignment

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **45. Front Suspension - Overhaul Strut Type**

Source:

[https://help.cccis.com/webhelp/motor/gte/Content/FRONTSUSPENSION\\_OVERHAULSTRUTTYPE.htm](https://help.cccis.com/webhelp/motor/gte/Content/FRONTSUSPENSION_OVERHAULSTRUTTYPE.htm)

## **Front Suspension - Overhaul Strut Type**

ABS sensor

Air suspension components

Ball joint

Bearing

Bearing seal

Brake rotor

Coil spring & components

Coil spring cap

Hub

Knuckle

Lower control arm

Lower control arm bushing

Overhaul procedure as defined

Raise & support vehicle (1 time)

Splash shield

Stabilizer bar end / link

Steering arm

Strut radius rod & bushing

Strut cartridge / assembly

Strut seat

Brake calipers R&I

Brake pad or shoes

Brakes adjust & bleed

Crossmember

Drivetrain components

Road wheel

Stabilizer bar

Steering linkage

Steering gear

Suspension / wheel alignment

Wheel cylinders

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **46. Front Suspension - Overhaul S.L.A. Type**

Source:

[https://help.cccis.com/webhelp/motor/gte/Content/FRONTSUSPENSION\\_OVERHAULS\\_L\\_A\\_TYPE.htm](https://help.cccis.com/webhelp/motor/gte/Content/FRONTSUSPENSION_OVERHAULS_L_A_TYPE.htm)

### **Front Suspension - Overhaul S.L.A. Type**

SPECIAL NOTATION:

S.L.A. is defined as short upper control arm and long lower control arm.

This includes Ball Joint and Torsion Bar Systems.

ABS sensor

Ball joint

Bearing seal R&R

Bearing

Brake rotor

Coil spring components

Hub

Knuckle

Lower control arm components

Overhaul procedure as defined

Radius rod & bushing D&R

Raise & support vehicle (1 time)

Shock absorber

Splash shield

Steering arm

Torsion bar

Upper control arm components

Axle or carrier assy (FWD or AWD)

Brake calipers R&I

Brake pad or shoes

Brakes adjust & bleed

Crossmember

Road wheel

Stabilizer bar

Steering gear or linkage

Suspension / wheel alignment

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **47. Fuel Tank**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/FUELTANK.htm>

### **Fuel Tank**

SPECIAL NOTATION:

Disconnect at fuel filler neck.

Fuel pump (if necessary)

Sending unit

Exhaust system

Fuel tank drain & refill

Suspension & drivetrain R&I

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **48. Fender - Front Rails**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/Fender\\_FrontRails.htm](https://help.cccis.com/webhelp/motor/gte/Content/Fender_FrontRails.htm)

### **Fender - Front Rails**

Adhesive application if required (all types)

Carpet & insulation turn back

Caulk / Seam sealer

Cowl trim

Cutting & welding as necessary

Grind, fill and smooth welded seams (up to 150 grit sandpaper)

Rivet removal and installation if required

Adjacent panels  
Battery  
Bolted on parts  
Bumper assembly  
Carpet, insulation or seat R&I  
Decals, labels  
Electrical wiring  
Energy absorber  
Engine  
Exhaust system  
Front suspension  
Outer sheet metal  
Pulling or straightening time  
Set up on frame machine & diagnosis  
Steering components  
Suspension / Wheel alignment  
Welded bracket & braces transfer  
Things to Consider  
Acoustical and Structural Foam Fillers  
Bolted On Parts  
Footnotes  
Included and/or Not Included Labor Operations  
Labor Time Listings  
Labor Time Premise  
R & I (Remove & Reinstall)  
Sectioning  
Special Substrate Metals  
Structural Component Identification  
Refinish Starting Point  
Basic Color Coat Application  
Refinish Time Listings  
Refinish Time Premise  
Refinish Processes  
Clear Coat Finishes (Base Coat / Clear Coat)  
Panel and/or Component Designation

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **49. Fender - Inner Panel (Welded Apron & Rails)**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/Fender\\_InnerPanel\\_Apron\\_Rails\\_.htm](https://help.cccis.com/webhelp/motor/gte/Content/Fender_InnerPanel_Apron_Rails_.htm)

### **Fender - Inner Panel (Welded Apron & Rails)**

#### **SPECIAL NOTATION:**

Detach and weld at floor pan, radiator support, hinge pillar & cowl panel.

Adhesive application if required (all types)

Battery tray

Carpet & insulation turn back

Caulk / seam sealer

Cowl trim

Cutting & welding as necessary

Grind, fill and smooth welded seams (up to 150 grit sandpaper)

Rivet removal and installation if required

Adjacent panels

Air inlet system (if necessary)

Attached standard items

Battery

Bolted on parts

Bumper assembly

Carpet, insulation or seat R&I

Cruise control

Dash panel

Decals, labels

Electrical wiring

Energy absorber

Engine

Front suspension assembly

Hood hinges & support cylinder

Horns

Outer sheet metal

Pulling or straightening time

Reservoir  
Set up on frame machine & diagnosis  
Steering components  
Suspension / wheel alignment  
Washer / Coolant reservoir  
Welded brackets & braces transfer  
Things to Consider  
Acoustical and Structural Foam Fillers  
Bolted On Parts  
Footnotes  
Included and/or Not Included Labor Operations  
Labor Time Listings  
Labor Time Premise  
R & I (Remove & Reinstall)  
Sectioning  
Special Substrate Metals  
Structural Component Identification  
Refinish Starting Point  
Basic Color Coat Application  
Refinish Time Listings  
Refinish Time Premise  
Refinish Processes  
Clear Coat Finishes (Base Coat / Clear Coat)  
Panel and/or Component Designation  
Things to Know  
Weld Zone / Adjacent Panel

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **50. Fender - Outer Panel**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/Fender\\_OuterPanel.htm](https://help.cccis.com/webhelp/motor/gte/Content/Fender_OuterPanel.htm)

### **Fender - Outer Panel**

## SPECIAL NOTATION:

Some replacement components may or may not be supplied with duplicated OEM caulk/seam sealer. This is not included in R&R time and requires an on-the-spot evaluation.

Align to vehicle

Fillers (if mounted to fender)

Cornering lamps (if mounted to fender)

Fender liner

Scoop

Side marker

Side repeater lamp (if mounted to fender)

Antenna

Bumper R&I

Battery

Emblems & nameplates

Grille

Header panel

Hood

Inner panels

Lamp aiming

Mirror

Moldings

Mud guard

Road wheel

Spoilers & flares

Stripe tape, decals or overlays

Things to Consider

Add If Required

Adhesive Type Moldings, Nameplates & Emblems

Bolt Or Clip-On Moldings

Damaged or Non-Reusable Moldings and Fasteners

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

R & I (Remove & Reinstall)

Special Substrate Metals

Refinish Starting Point

Basic Color Coat Application  
Parts Supplied In Color  
Refinish Clear Coat Application  
Refinish Time Listings  
Refinish Time Premise  
Refinish Processes  
Clear Coat Finishes (Base Coat / Clear Coat)  
Color Blend (Adjacent Panels)  
Deductions to Basic Refinish Times (Refinish Overlap)  
Panel and/or Component Designation  
Two-Tone Refinishing (Second Color Tone Application)  
Things to Know  
Bagging (Cover Entire Vehicle Exterior)  
Clear Coat Undamaged Panel  
De-Nib & Polish  
Edges Of New Parts (Edging)  
Stone Chip Guard (Protective Material)  
Three-Stage Color Blend (Adjacent Panels)  
Three-Stage Finishes (Base / Mica / Clear Coat)  
Wet / Dry Sand, Rub-Out & Buff

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **51. Floor Pan - Front & Center Floor Pan**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/FloorPan\\_Front\\_Center\\_Floor\\_Pan.htm](https://help.cccis.com/webhelp/motor/gte/Content/FloorPan_Front_Center_Floor_Pan.htm)

### **Floor Pan - Front & Center Floor Pan**

Adhesive application if required (all types)

Cutting & welding as necessary

Caulk / seam sealer

Rivet removal and installation if required

- Bolted on parts
- Electrical wiring

- Exhaust system
- Sound deadening pad

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **52. Footnotes**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/Footnotes.htm>

### **Footnotes**

Footnotes are used to present information specific to the indicated component or operation. Footnotes are not intended to be used in place of service repair information.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **53. Frame Machine Set-Up**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/Frame Machine Set up.htm>

### **Frame Machine Set-Up**

Due to the different types of frame machines used in the collision repair industry, labor times for frame machine set-up are not developed by MOTOR, nor otherwise included in any operation. Each frame machine manufacturer may have its own unique configurations and setup processes. For example, some machines are of a "drive-on" type while others are of a "dedicated bench" type, and there are procedural differences between the two set-up methods. Additionally, there may be variables unique to the actual vehicle that may increase or decrease frame machine set-up time. MOTOR suggests using an on-the-spot evaluation to determine an appropriate frame machine set-up time.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **54. Front Bumper Cover - Overhaul Cover Type**

## Front Bumper Cover - Overhaul Cover Type

### SPECIAL NOTATION:

The removal of mold-release agents from and preparation/priming of unprimed bumpers is required and is NOT INCLUDED. Where the OEM has provided information regarding the availability of unprimed bumper covers, an "Add if Required" labor operation has been added to the bumper group.

Align to vehicle

Disassemble cover assembly

Emblem & nameplate

Energy absorber (when mounted to cover)

Grille (when mounted to bumper)

Lamps (when mounted to bumper)

Molding & impact strip

Reassemble unit

Reflectors

Replace damaged parts

Tow eye caps

Valance panel (if mounted to bumper)

Aim lamps

Air bag sensor

Battery

Bumper brackets/frame rail extensions (if mounted to frame rail)

Drill time

Distance Sensor

Lamp aiming

Lamps (optional equipment or not mounted to bumper)

License plate/bracket

Reinforcement/impact bar (if mounted to frame rail)

Stripe tape, decals or overlays

Valance panel / spoiler (when not mounted to bumper)

Things to Consider

Adhesive Type Moldings, Nameplates & Emblems

Damaged or Non-Reusable Moldings and Fasteners

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

Overhaul

Part Quantities

R & I (Remove & Reinstall)

Refinish Starting Point

Basic Color Coat Application

Parts Supplied In Color

Refinish Clear Coat Application

Refinish Time Listings

Refinish Time Premise

Unprimed Flexible Component Preparation

Refinish Processes

Clear Coat Finishes (Base Coat / Clear Coat)

Panel and/or Component Designation

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **55. Front Bumper Assembly - R & I All Types**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/FrontBumperAssembly\\_R\\_IAllTypes.htm](https://help.cccis.com/webhelp/motor/gte/Content/FrontBumperAssembly_R_IAllTypes.htm)

### **Front Bumper Assembly - R & I All Types**

Align to vehicle

Face bar / bumper cover assembly R&I

Air bag sensor

Battery

Emblems & nameplates

Energy Absorber, all types

Lamp aiming

Lamps (when not mounted in bumper)

Moldings & impact strip

Stripe tape, decals or overlays

Valance panel / spoiler (when not mounted to bumper)

Things to Consider

Damaged or Non-Reusable Moldings and Fasteners

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

R & I (Remove & Reinstall)

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **56. Front Bumper - R & R Face Bar Type**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/FrontBumperAssembly\\_R\\_RFaceBarType.htm](https://help.cccis.com/webhelp/motor/gte/Content/FrontBumperAssembly_R_RFaceBarType.htm)

### **Front Bumper - R & R Face Bar Type**

Align to vehicle

Emblem & nameplate

Face bar R&I

Guards

Guard cushions

Lamps (when mounted to bumper)

Molding & impact strip

Air bag sensor

Battery

Distance sensor

Energy absorber if mounted to frame rail (all types)

Lamp aiming

Lamps (optional equipment, or not mounted to bumper)

License plate/bracket

Stripe tape, decals or overlays

Valance panel / spoiler (when not mounted to bumper)

Things to Consider

Add If Required

Adhesive Type Moldings, Nameplates & Emblems

Damaged or Non-Reusable Moldings and Fasteners

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

Part Quantities

R & I (Remove & Reinstall)

Refinish Starting Point

Basic Color Coat Application

Parts Supplied In Color

Refinish Clear Coat Application

Refinish Time Listings

Refinish Time Premise

Refinish Processes

Clear Coat Finishes (Base Coat / Clear Coat)

Panel and/or Component Designation

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **57. Front Bumper - R & R Cover Type**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/FrontBumperAssembly\\_R\\_RFasciaType.htm](https://help.cccis.com/webhelp/motor/gte/Content/FrontBumperAssembly_R_RFasciaType.htm)

### **Front Bumper - R & R Cover Type**

#### **SPECIAL NOTATION:**

The removal of mold-release agents from and preparation/priming of unprimed bumpers is required and is NOT INCLUDED. Where the OEM has provided information regarding the availability of unprimed bumper covers, an "Add if Required" labor operation has been added to the bumper group.

Align to vehicle

Emblem & nameplate

Energy absorber (when mounted to cover)

Front bumper cover R&R

Grille s (when mounted to bumper)

Lamps (when mounted to bumper)

Molding & impact strip

Reflectors

Tow eye caps

Valance panel (if mounted to bumper)

Aim lamps

Air bag sensor

Battery

Bumper brackets/frame rail extensions (if mounted to frame rail)

Distance sensor

Energy absorber (if mounted to frame rail)

Lamps (optional equipment or not mounted to bumper)

License plate/bracket

Reinforcement/impact bar

Stripe tape, decals or overlays

Valance Panel / Spoiler (when not mounted to bumper)

Things to Consider

Add If Required

Adhesive Type Moldings, Nameplates & Emblems

Damaged or Non-Reusable Moldings and Fasteners

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

Part Quantities

R & I (Remove & Reinstall)

Refinish Starting Point

Basic Color Coat Application

Parts Supplied In Color

Refinish Clear Coat Application

Refinish Time Listings

Refinish Time Premise

Unprimed Flexible Component Preparation

Refinish Processes

Clear Coat Finishes (Base Coat / Clear Coat)

Panel and/or Component Designation

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## 58. Front Suspension - Overhaul I-Beam / Tube Type

Source: [https://help.cccis.com/webhelp/motor/gte/Content/FrontSuspension\\_OverhaulTwinI\\_BeamType.htm](https://help.cccis.com/webhelp/motor/gte/Content/FrontSuspension_OverhaulTwinI_BeamType.htm)

### Front Suspension - Overhaul I-Beam / Tube Type

ABS sensor

Axle beam / tube

Ball joints

Bearing seal R&R

Bearing

Brake rotor

Coil spring seat

Coil spring

Coil spring insulator

Control arms

Hub

Knuckle / spindle

Overhaul procedure as defined

Raise & support vehicle (1 time)

Radius rod & bushing

Shock absorber

Splash shield

Steering arm

Axle or carrier assy (4WD) R&I

Brake caliper R&I

Brake pad or shoes

Brakes adjust & bleed

Drive train parts

Road wheel

Stabilizer bar

Steering gear or linkage

Suspension / wheel alignment

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## 59. Grille

Source: <https://help.cccis.com/webhelp/motor/gte/Content/GRILLE.htm>

### Grille

Emblems & nameplates

Grille

Lamp (in grille)

Moldings & ornamentation

Adjacent panels or components

Camera mounted to grille

Lamp aiming

Stripe tape, decals

Things to Consider

Add If Required

Adhesive Type Moldings, Nameplates & Emblems

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

Refinish Starting Point

Basic Color Coat Application

Parts Supplied In Color

Refinish Clear Coat Application

Refinish Time Listings

Refinish Time Premise

Refinish Processes

Clear Coat Finishes (Base Coat / Clear Coat)

Deductions to Basic Refinish Times (Refinish Overlap)

Panel and/or Component Designation

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **60. MOTOR GUIDE TO ESTIMATING rev 11-25**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/GeneralInformation.htm>

## **MOTOR GUIDE TO ESTIMATING rev 11-25**

### **### General Information Overview**

Operation times reported herein are compiled from available manufacturer data, as well as our own evaluation of shop data, and are published only as an estimating guide.

This Guide To Estimating provides the guidelines that apply to MOTOR Crash Estimating Data. Knowledge and application of these guidelines will assist the estimator in developing a clear estimate, reflecting as accurately as possible the requirements to perform operations listed. To maintain accuracy, estimators must frequently refer to these pages throughout the estimating process.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **61. Hazardous Materials**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/HAZARDOUSMATERIALS.htm>

## **Hazardous Materials**

Regulations and safety considerations may require protective clothing, respirator, protective eyeglasses, or ear plugs be worn in certain areas of collision repair facilities. Caution must be exercised when dealing with isocyanates, flammables, fumes, liquids, and airborne particles during the repair process as well as any subsequent cleanup of vehicle and work location as hazardous materials may be present

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **62. Header Panel or Headlamp Mounting Panel**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/HEADERPANELORFRONTCOVER.htm>

## **Header Panel or Headlamp Mounting Panel**

Align to vehicle

Bolt on extensions & fillers

Emblems & nameplates

Grille

Lamp

Moldings

Ornamentation

Bumper assembly

Lamp aiming

Stripe tape, decals or overlays

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **63. Heater**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/HEATER.htm>

### **Heater**

R&R time is listed with parts

Cost of coolant

Drain and refill fluids

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **64. High Voltage Battery**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/HIGHVOLTAGEBATTERY.htm>

### **High Voltage Battery**

Battery removal & installation

Disconnect & reconnect high voltage battery

Drain and refill coolant

Purge air from system (one time)

Skid plates

Splash shields

Under covers

Deactivate/activate high voltage system

Individual cell replacement or testing

Inspection of personal protective equipment

Interior trim

Set up of lifting table or device

Suspension components

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **65. Hood**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/HOOD.htm>

### **Hood**

#### **SPECIAL NOTATION:**

Some replacement components may or may not be supplied with duplicated OEM caulk / seam sealer, this is not included in R&R time and requires an on-the-spot evaluation.

Disconnect at hinges.

Air inlet system (if necessary)

Align to vehicle

Grille (if attached)

Hood lamp

Hood panel R&R

Insulation

Scoop

Safety catch

Striker

Washer hoses & nozzle (mounted to hood)

Weatherstrips & seals (mounted on hood)

Emblems & nameplates

Hinge

Hood lock

Moldings & ornamentation

Stripe tape, decals or labels

Things to Consider

Add If Required

Adhesive Type Moldings, Nameplates & Emblems

Drilling Operations

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

R & R (Remove & Replace)

Refinish Starting Point

Basic Color Coat Application

Parts Supplied In Color

Refinish Clear Coat Application

Refinish Time Listings

Refinish Time Premise

Refinish Processes

Clear Coat Finishes (Base Coat / Clear Coat)

Color Blend (Adjacent Panels)

Deductions to Basic Refinish Times (Refinish Overlap)

Panel and/or Component Designation

Underside of Hoods, Lids or Gates

Things to Know

Bagging (Cover Entire Vehicle Exterior)

Clear Coat Undamaged Panel

De-Nib & Polish

Edges Of New Parts (Edging)

Three-Stage Color Blend (Adjacent Panels)

Three-Stage Finishes (Base / Mica / Clear Coat)

Wet / Dry Sand, Rub-Out & Buff

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## 66. Hybrid/Electric Vehicles

Source: [https://help.cccis.com/webhelp/motor/gte/Content/HYBRID\\_VEHICLES.htm](https://help.cccis.com/webhelp/motor/gte/Content/HYBRID_VEHICLES.htm)

### Hybrid/Electric Vehicles

#### Warning High Voltage Electrical System!

These vehicles are equipped with high voltage batteries and are capable of causing electrical shock. Failure to use caution in the proper disconnection of this electrical system may result in serious injury or death. For safety and ease of identification, electrical cables carrying the high voltage are colored orange. Any orange-colored cable should be avoided until the high voltage electrical system has been disabled. Follow OEM safety precautions including wearing personal protective equipment rated for working on Hybrid/Electric Vehicles. Follow OEM service repair information for disabling the high voltage electrical system before beginning any repairs.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## 67. Instrument Panel

Source: <https://help.cccis.com/webhelp/motor/gte/Content/INSTRUMENTPANEL.htm>

### Instrument Panel

#### SPECIAL NOTATION:

Air bag considerations are included in all Instrument Panel operations unless " Add If Required " heading is listed in text.

Crash pad R&I

Display unit R&I

Emblems & nameplates

Glove box R&I

Heater & A/C controls

Instrument cluster R&I

Lamps & switches

Metal reinforcement

Moldings & ornamentation R&I

Radio

Side covers

Speakers

Console

Inner cowl support braces & brackets

Remote control mirrors

Front Seat R&I (if necessary)

Steering column

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **68. Included and/or Not Included Labor Operations**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/IncludedOperations.htm>

### **Included and/or Not Included Labor Operations**

#### **INCLUDED OPERATIONS:**

When items or operations appear in the Guide To Estimating Pages under the "Included" heading it means that the operation is performed in conjunction with another operation. For example, Steering Wheel R&I is an individual operation, but when replacing a steering column, steering wheel R&I is also performed and therefore included in Steering Column R&R.

If an item is listed without a qualifier, it means all labor has been considered within the indicated repair procedure. If a specific qualifier (such as R&I) appears, it means only the specified qualifier applies.

#### **NOT INCLUDED OPERATIONS:**

Items or operations listed under "Does Not Include" were not considered in the development of published estimated work times. These operations may, or may not, be required depending upon the vehicle or repair process used. If any of these items or operations are required, they should be considered by the estimator. If a specific qualifier (such as R&I) appears, it means only the specified qualifier applies.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **69. Estimated Work Time Premise**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/LABORTIMEPREMISE.htm>

## Estimated Work Time Premise

The estimated work times reported in this publication are to be used as a GUIDE ONLY. MOTOR recommends a review of the OE service information at the time of estimate preparation to confirm the OE manufacturer repair procedure(s). MOTOR assumes the repair facility is equipped with the necessary tools, equipment, and training to complete any given repair following published vehicle manufacturer repair procedures. Subletting repairs must be considered a business decision by the repair facility and does not change declared inclusions stated in each applicable Guide To Estimating section. Reported estimated work times include one pre-fitment of OEM exterior body panels, body structure components and bumpers, which includes normal adjustment procedures within the range of motion permitted by the fastener(s) or component to ensure proper placement of the individual new component being replaced. Any additional fitment of a component would not be included and would require an on-the-spot evaluation. Pre-fitment would not be applicable to remanufactured or aftermarket components. Pre-fitment of R&I components is not included.

Prebuild/mockup of multiple/adjacent panels to verify the dimensional accuracy of adjoining parts is not included. Reported times include tube/paddled OEM caulking and seam sealer removal/application on bonded, riveted, and welded replacement panels. Spray-able seam sealer equipment requires preparation and adjustment before application and is NOT INCLUDED IN ESTIMATED WORK TIME. MOTOR estimated work times are for the actual time that the technician spends on the physical procedure required to complete repairs of the damage. Please note: Wait time i.e., drying, curing is not considered with MOTOR estimated work times.

Estimated work time(s) for a component attached with adhesives, welds, and/or rivets includes removal of rivets, welds, weld nugget(s), remaining adhesive and component removal. Prepping attachment surface and replacement component(s), drilling as required, resizing of rivet holes, deburring, Pyrosil & adhesive application, EMC screws (if required), installing rivets, and welding as necessary.

Estimated work times do not apply to vehicles with equipment other than that supplied by the vehicle manufacturer as standard or regular production options. If other equipment is used, the time may be adjusted to compensate for the variables. Removal and replacement of exchanged or used parts is not considered. If additional aligning or repair must be made, such factors should be considered when developing the estimate. Items not listed under the INCLUDED/DOES NOT INCLUDE heading for any given procedure have not been considered in the estimated work time development for that procedure, unless specified by a footnote. All included/not included items for labor procedures listed are for component R&R and R&I procedures unless otherwise indicated in operation heading.

Many OEMs recommend a pre-and post-repair diagnostic scan on all vehicles involved in a collision that could reveal pre-accident or accident-related damage. Due to the variety of vehicle manufacturer approved scan tools, both OEM and aftermarket, as well as the variation from vehicle to vehicle and repair to repair, MOTOR is unable to develop a time that accounts for these different scenarios. Therefore, MOTOR recommends an on-the-spot evaluation to determine an appropriate estimated time to perform diagnostic scans for each repair. Please refer to OEM position statements and repair procedures for more information.

OPERATION TIMES LISTED ARE BASED ON NEW UNDAMAGED PARTS INSTALLED ON NEW UNDAMAGED VEHICLES AS INDIVIDUAL OPERATIONS. TIME HAS NOT BEEN CONSIDERED FOR ALIGNMENT PULLS, DAMAGE RELATED ACCESS TIME, DAMAGED, USED, REMANUFACTURED OR AFTERMARKET PARTS. SOME OPERATION TIMES ARE APPLICABLE AFTER BOLTED, ATTACHED OR RELATED PARTS HAVE BEEN REMOVED. REFER TO SPECIFIC FOOTNOTES ATTACHED TO OPERATION TIME LISTING.

ESTIMATED WORK TIME DOES NOT INCLUDE:

SPECIAL NOTATION:

The items listed below apply to all labor procedures. If an estimated work time is not available for the items below, an on-the-spot evaluation is recommended to determine an appropriate estimated time.

- A/C System, Evacuate and Recharge
- Access to repair information/subscription cost
- Aftermarket & OEM accessories
- Alignment, straightening, or verifying the dimensional accuracy of related parts
- Alignment check of front or rear suspension / steering
- Anticorrosion material restoration / application
- Battery D&R / recharge
- Brackets & braces transfer
- Brakes, bleed and adjust
- Broken glass removal or clean up
- Capping, covering electrical connectors and fluid lines (all types) to prevent contamination
- Caulk (non-OEM), sound insulate or paint inner areas
- Clean up or detailing of vehicle prior to delivery
- Computer control module D&R / relearn
- Conversion Vans (special components, equipment and trim)
- Cutting, pulling or pushing collision damaged parts for access
- Damaged or defective replacement parts
- Drain & refill fuel tank
- Drilling, modification or fabrication of mounting holes
- Fabricate templates , reinforcing inserts, sleeves or flanges
- Filling, plugging and finishing of unneeded holes in new parts
- Information label installation
- Maintenance costs of building or equipment
- Material /consumable product costs
- Pinch weld clamp damage repair
- Refinishing
- Repair information retrieval/lookup
- Reset electronic memory functions after battery disconnect
- Rivet gun set-up or preparation
- Road test vehicle
- Rusted, frozen, broken or corrosion damaged components or fasteners
- Scaffolding/temporary support setup and removal
- Scan tool clear / reset electronic module
- Scan tool diagnostics
- Second or subsequent fitment of a component to the vehicle, for any reason (including verification of dimensional accuracy or adjacent part alignment)
- Steering Angle Sensor Recalibration

- Straighten or align used, reconditioned or non-OEM parts
- Structural damage diagnosis and vehicle set up time
- Structural foam removal or application
- Test for water leaks (except Windshield, Back Glass, and Stationary Glass)
- Test panel / Spray caulk
- Testing/practice to duplicate OE caulking appearance
- Undercoating, tar or grease removal
- Unprimed bumpers, removal of mold-release agents
- Waste disposal fees (all types)
- Weld through primer / cost & application
- Welded seam surface finishing finer than 150 grit sandpaper
- Welder set-up or preparation
- Wheel or hub cap locks R&I

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## 70. Liftgate

Source: <https://help.cccis.com/webhelp/motor/gte/Content/LIFTGATE.htm>

### Liftgate

#### SPECIAL NOTATION:

Some replacement components may or may not be supplied with duplicated OEM caulk / seam sealer. This is not included in R&R time and requires an on-the-spot evaluation.

Disconnect wiring at connector if accessible.

Disconnect at hinge & support cylinder.

Align to vehicle

Applique panel

Bonded glass installation

Electrical wiring (standard features)

Glass (moveable)

Glass moldings

Finish panel

Finish plate

Finish molding

Finish trim

Inside release

Interior trim

Internal mechanisms

Lamps

Latch or striker attached to gate

License plate (if attached)

License pocket

Lift gate switch (exterior)

Lock & cylinder assembly (if necessary)

Outside handle

Trim moldings

Trim panel R&I

Watershield

Weatherstrips (if attached to lift gate)

Window / run channel

Back glass wiper or washer

Bonded glass removal from damaged liftgate

Camera

Electrical wiring (if not detachable from main body harness)

Emblems & nameplates

Hinges

Lift gate switch (interior)

Lock cylinder recoding

Moldings (except as defined in Included)

Pinch sensors

Spare tire carrier

Special interior trim

Spoiler, flare or louver

Stripe tape, decals or overlays

Things to Consider

Add If Required

Adhesive Type Moldings, Nameplates & Emblems

Damaged or Non-Reusable Moldings and Fasteners

Decals, Stripe Tape & Overlays

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

R & I (Remove & Reinstall)

Special Substrate Metals

Structural Glass

Refinish Starting Point

Basic Color Coat Application

Parts Supplied In Color

Refinish Clear Coat Application

Refinish Time Listings

Refinish Time Premise

Refinish Processes

Clear Coat Finishes (Base Coat / Clear Coat)

Color Blend (Adjacent Panels)

Deductions to Basic Refinish Times (Refinish Overlap)

Panel and/or Component Designation

Two-Tone Refinishing (Second Color Tone Application)

Underside of Hoods, Lids or Gates

Things to Know

Bagging (Cover Entire Vehicle Exterior)

Clear Coat Undamaged Panel

De-Nib & Polish

Edges Of New Parts (Edging)

Stone Chip Guard (Protective Material)

Three-Stage Color Blend (Adjacent Panels)

Three-Stage Finishes (Base / Mica / Clear Coat)

Wet / Dry Sand, Rub-Out & Buff

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **71. Estimated Work Times**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/LaborTimeListings.htm>

## Estimated Work Times

All operation times are listed in hours and tenths of an hour. A time listed as 3.5 indicates three and one-half hours.

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## 72. Front Lamps

Source: <https://help.cccis.com/webhelp/motor/gte/Content/Lamps.htm>

### Front Lamps

#### COMPOSITE TYPE

Ballast

Bulb(s)

Circuit Board

Ignitor

Modules

Motors

Lamp aiming

Headlamp door / bezel

#### SEALED BEAM TYPE

Mount ring

Retainer ring

Seal beam

Lamp aiming

Headlamp door / bezel

Rear Lamps

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## 73. NAGS - Glass Prices

Source: [https://help.cccis.com/webhelp/motor/gte/Content/NAGS\\_GlassPrices.htm](https://help.cccis.com/webhelp/motor/gte/Content/NAGS_GlassPrices.htm)

## **NAGS - Glass Prices**

Glass Prices: MOTOR includes, when available, both the vehicle manufacturer and the NAGS benchmark prices as applicable to each make and model. All NAGS part numbers and benchmark prices are provided from National Auto Glass Specifications, a division of Mitchell International, Inc. Estimated work times listed on the line with the NAGS information are MOTOR suggested estimated work times ; NAGS labor operation times are not included.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **74. OEM Collision Repair Network**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/OEM\\_COLLISION\\_REPAIR\\_NETWORK.htm](https://help.cccis.com/webhelp/motor/gte/Content/OEM_COLLISION_REPAIR_NETWORK.htm)

### **OEM Collision Repair Network**

Some OEM(s) manufacture aluminum vehicles that have an established repair network for their aluminum vehicles and require collision repairs to be performed at an OEM authorized aluminum repair center. Refer to the OEM for complete information on collision repair network program.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **75. OEM Component Replacement - General**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/OEM\\_COMPONENT\\_REPLACEMENT-GENERAL.htm](https://help.cccis.com/webhelp/motor/gte/Content/OEM_COMPONENT_REPLACEMENT-GENERAL.htm)

### **OEM Component Replacement - General**

The collision repair facility must have the equipment, training, and experience using the latest factory/industry information and procedures. Ultimately, the safety and quality of any repair will depend not only on the equipment that is used and the procedure that is followed, but also on the skill and knowledge of the repair technicians, the steps they take to control repair quality and how well they check the details of the repair.

Some repair operations reported in this guide may be beyond the capability of the repair shop in terms of equipment, skill and knowledge, etc. In this case the repair shop should not attempt the repair.

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Extreme care and caution should always be used when performing collision repairs on any vehicle. All repairs must be of the highest quality using the latest factory recommended repair procedures with the following concerns kept in mind:

- **Passenger Protection:** Replacing damaged parts of a car designed to crush in a collision may reduce occupant protection in a future collision if the structural integrity is not restored.
- **Safety & Handling:** An improper repair can create braking, handling, vibration and noise problems. This may lead to possible safety problems.
- **Corrosion Protection:** A vehicle's ability to resist corrosion may be reduced if improper repairs are made and the manufacturer's corrosion protection is not restored. This may lead to future safety and/or resale value problems.
- **Resale Value:** The resale value of a vehicle can be reduced dramatically by improper repairs which may lead to a financial loss to the vehicle owner.

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## **76. Overhaul**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/OVERHAUL.htm>

### **Overhaul**

Remove an assembly from the vehicle, disassemble, clean, inspect, replace parts as needed, reassemble, install and adjust (except wheel and suspension alignment). Overhaul time should be used only if the time for individual parts (less overlap) is more than the overhaul time. Overhaul operations include component R&R procedure steps and inclusions unless otherwise noted.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **77. Overlap**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/OVERLAP.htm>

### **Overlap**

When replacing two or more components the duplication of included labor procedures is known as overlap. Labor procedures (R&R / R&I / D&R) that create overlap include but are not limited to, mechanical attachment, welding, bonding and/or technician preparation.

For example, when replacing a quarter panel and rear body panel on the same vehicle, the common required labor procedures necessary to remove and replace or reinstall these components is known as overlap.

When a labor overlap condition exists, less time is required to replace adjoining components collectively than is required when they are replaced individually.

Overlap labor information is generally included at the beginning of each group within each chapter.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **78. O.E.M. (Original Equipment Manufacturer)**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/O\\_E\\_M\\_\\_OriginalEquipmentManufacturer\\_.htm](https://help.cccis.com/webhelp/motor/gte/Content/O_E_M__OriginalEquipmentManufacturer_.htm)

### **O.E.M. (Original Equipment Manufacturer)**

Used to define original vehicle manufacturer.

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## **79. Panel Bonding (Metal Adhesives)**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/PANEL\\_BONDING.htm](https://help.cccis.com/webhelp/motor/gte/Content/PANEL_BONDING.htm)

### **Panel Bonding (Metal Adhesives)**

OEM labor procedures for "Panel bonding" requires different procedural steps versus welding, i.e.; panel / vehicle preparation, sectioning insert / sleeve, additional panel fit, and adhesive application unless otherwise specified in a footnote attached to the sectioning operation. MOTOR published welded panel replacement labor times may be applied for "Panel bonding" and would include all the necessary procedure steps as outlined by the OEM. Refer to the Guide To Estimating pages for specific INCLUDED / NOT INCLUDED operations, an adjustment in time may or may not apply after an on-the-spot evaluation of required procedures. Refer to OEM for specific repair recommendations / procedures and replacement product(s).

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **80. Part Prices**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/PARTPRICES.htm>

### **Part Prices**

Prices used in MOTOR Crash Estimating Guides are factory suggested list prices, as supplied to us by the manufacturer. Dealers may not be required to sell at this price, and may be free to determine their own prices. When a price is prefixed by the letter "D" it indicates that this part has been discontinued by the manufacturer and for your convenience, we continue to publish the last available price. When a part is prefixed by the letter "C", this indicates a core value assigned by the manufacturer has been deducted from the manufacturer's suggested retail price. When a price is prefixed by the letter "A" it indicates a Special Marketing Program part.

General Motors Target Marketing Program: This program has been discontinued by GM. Parts indicated with the letter "A" represent components included in the program. These parts do not have a suggested list price supplied by General Motors. The price shown following the letter "A" is an approximation which you may find helpful. These prices are to be regarded as a GUIDE only. The actual price, which may be higher or lower than the published price, can be determined by contacting the General Motors dealer in your area.

Mopar Competitive Crash Parts Program: The prices for these parts may carry a special discount. Actual price should be determined by calling the dealer in your local area. Dealers are not required to participate in this program, and are free to determine their own prices for these parts.

Part prices do not include cost of state and local taxes, bolts, rivets, screws, nuts, washers, clips, fasteners, paint / body repair materials and refinishing unless otherwise noted.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **81. Part Quantities**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/PART\\_QUANTITIES.htm](https://help.cccis.com/webhelp/motor/gte/Content/PART_QUANTITIES.htm)

### **Part Quantities**

Many procedures require replacement of parts such as fasteners (bolts, rivets, screws, nuts, clips, for example) or washers. The applicable quantities for these parts are provided whenever possible. This information will be provided as reported by the OEM.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **82. PICKUP TRUCK – BOX ASSEMBLY R&I**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/PICKUPTRUCKBOXASSEMBLY\\_R\\_I.htm](https://help.cccis.com/webhelp/motor/gte/Content/PICKUPTRUCKBOXASSEMBLY_R_I.htm)

### **PICKUP TRUCK – BOX ASSEMBLY R&I**

#### **SPECIAL NOTATION:**

Pick-up box R&I is considered by MOTOR to be the removal of the fasteners securing the box to the frame of the vehicle, D&R necessary wiring, D&R fuel neck and removing the box from the vehicle and reinstalling to the vehicle. This operation is generally used for accessing the cab back or side panels or welds on the quarter panel.

Disconnect and remove box

Reinstall and reconnect box

Accessory items

Bumper R&I

Fuel door

Fuel tank

Lamps

Tailgate R&I

Things to Consider

Acoustical and Structural Foam Fillers

Add If Required

Adhesive Type Moldings, Nameplates & Emblems

Damaged or Non-Reusable Moldings and Fasteners

Decals, Stripe Tape & Overlays

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

Panel Bonding (Metal Adhesives)

R & I (Remove & Reinstall)

Sectioning

Special Substrate Metals

Refinish Starting Point

Basic Color Coat Application

Parts Supplied In Color  
Refinish Clear Coat Application  
Refinish Time Listings  
Refinish Time Premise  
Refinish Processes  
Clear Coat Finishes (Base Coat / Clear Coat)  
Color Blend (Adjacent Panels)  
Deductions to Basic Refinish Times (Refinish Overlap)  
Panel and/or Component Designation  
Two-Tone Refinishing (Second Color Tone Application)  
Things to Know  
Bagging (Cover Entire Vehicle Exterior)  
Clear Coat Undamaged Panel  
De-Nib & Polish  
Edges Of New Parts (Edging)  
Stone Chip Guard (Protective Material)  
Three-Stage Color Blend (Adjacent Panels)  
Three-Stage Finishes (Base / Mica / Clear Coat)  
Weld Zone / Adjacent Panel  
Wet / Dry Sand, Rub-Out & Buff

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## **83. Pickup Truck - Uniside with Side or Corner Panel**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/PICKUP\\_TRUCK\\_UNISIDE\\_WITH\\_SIDE\\_OR\\_CORNER\\_PANEL.htm](https://help.cccis.com/webhelp/motor/gte/Content/PICKUP_TRUCK_UNISIDE_WITH_SIDE_OR_CORNER_PANEL.htm)

### **Pickup Truck - Uniside with Side or Corner Panel**

#### **SPECIAL NOTATION:**

When replacing welded door hinges, add one half of the Door Hinge R&R time to Uniside R&R time.

Adhesive application if required (all types)

Back glass and moldings

Back panel trim  
Carpet & insulation turn back  
Caulk / seam sealer  
Center pillar trim (4 door models)  
Cutting and welding as necessary  
Door striker  
Front seat R&I  
Grind, fill & smooth welded seams (up to 150 grit sandpaper)  
Hinge pillar trim  
Jamb switches  
Rear seat R&I  
Rivet removal and installation if required  
Seat belt anchor  
Side glass & moldings  
Side panel trim  
Sill plates  
Weatherstrips  
Body side moldings  
Box set back  
Carpet & insulation R&I  
Control cables  
Doors  
Door hinge  
Electrical wiring  
Emblems & nameplates  
Fender  
Fluid lines  
Foam  
Headliner  
Instrument panel  
Road wheel  
Stripe tape, decals or overlay  
Windshield glass and moldings  
Things to Consider  
Acoustical and Structural Foam Fillers  
Add If Required  
Adhesive Type Moldings, Nameplates & Emblems

Damaged or Non-Reusable Moldings and Fasteners  
Decals, Stripe Tape & Overlays  
Footnotes  
Included and/or Not Included Labor Operations  
Labor Time Listings  
Labor Time Premise  
Panel Bonding (Metal Adhesives)  
R & I (Remove & Reinstall)  
Sectioning  
Special Substrate Metals  
Refinish Starting Point  
Basic Color Coat Application  
Parts Supplied In Color  
Refinish Clear Coat Application  
Refinish Time Listings  
Refinish Time Premise  
Refinish Processes  
Clear Coat Finishes (Base Coat / Clear Coat)  
Color Blend (Adjacent Panels)  
Deductions to Basic Refinish Times (Refinish Overlap)  
Panel and/or Component Designation  
Two-Tone Refinishing (Second Color Tone Application)  
Things to Know  
Bagging (Cover Entire Vehicle Exterior)  
Clear Coat Undamaged Panel  
De-Nib & Polish  
Edges Of New Parts (Edging)  
Stone Chip Guard (Protective Material)  
Three-Stage Color Blend (Adjacent Panels)  
Three-Stage Finishes (Base / Mica / Clear Coat)  
Weld Zone / Adjacent Panel  
Wet / Dry Sand, Rub-Out & Buff

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## 84. Panel and/or Component Designation

Source: [https://help.cccis.com/webhelp/motor/gte/Content/PanelAnd\\_OrComponentDesignation.htm](https://help.cccis.com/webhelp/motor/gte/Content/PanelAnd_OrComponentDesignation.htm)

### Panel and/or Component Designation

#### MAJOR PANELS / COMPONENTS

All panels or components with a base refinish time of 1.0 hour or greater are generally considered by MOTOR to be major panels. Example; grille header panel, fender, hood, cowl top panel, doors, roof panel, rocker panel, quarter panel, engine lid, trunk lid, lift gate, rear gate, rear body panel, truck cab corner and back panel, truck bed front and side panel, van side and corner panel.

#### MINOR PANELS / COMPONENTS

A panel or component with a base refinish time of less than 1.0 hour is a Minor Component. A minor panel(s) or component(s) are based on the panel or component being refinished in conjunction with a major panel. Therefore, retrieving color information, including paint chip info, mix paint, load sprayer, and clean sprayer would not be included when a minor panel would be refinished independent of a major panel.

#### FLEXIBLE PANELS / COMPONENTS

All panels or components for which paint systems require a flex agent added to the paint mix. Example: fascia covers, fillers, extensions, spoilers, etc.

Bumpers and other flexible components can only receive a maximum refinish overlap deduction of 0.2 hour.

No overlap deduction should be taken when calculating refinish time for a single item from this category.

A combination of items from this category refinished during a single, continuous procedure should be subject to the appropriate "Adjacent" or "Non-Adjacent" overlap formula deduction.

When a flex agent, or a separate paint mix procedure is not required, and when the flexible component is refinished during the same procedure with major or minor components, then flexible components should be considered the same designation as major or minor components for the purpose of calculating refinish overlap deductions, and/or multi-stage refinishing additions.

#### INDIVIDUAL PROCEDURE ITEMS / AREAS

Areas of a panel or component that are part of a main component, but are refinished during a procedure separate from the main component. Example: edges, jams, hinges, inside panels and the underside of hoods, deck lids, liftgates, etc.

No overlap deduction should be taken when calculating refinish time for items from this category.

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## 85. Parts Supplied In Color

Source: <https://help.cccis.com/webhelp/motor/gte/Content/PartsSuppliedInColor.htm>

## Parts Supplied In Color

Some replacement components such as instrument panels, trim panels and moldings can be ordered in various colors, and therefore may not require painting. In some instances specific colors may no longer be available from the manufacturer and therefore would require painting. MOTOR recommends the availability of appropriate replacement colors be confirmed before finalizing any estimate of repairs. Refinishing time should be added if needed color is not available.

NOTE: Air bag components should never be refinished. Refinishing these components will alter the original design and change deployment characteristics.

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## 86. Restraint System (Seat Belt)

Source: [https://help.cccis.com/webhelp/motor/gte/Content/PassengerRestraintSystem\\_SeatBelt\\_ShoulderHarness\\_.htm](https://help.cccis.com/webhelp/motor/gte/Content/PassengerRestraintSystem_SeatBelt_ShoulderHarness_.htm)

### Restraint System (Seat Belt)

Many automobile manufacturers recommend seat belt components be replaced when subjected to stress by occupants in a collision. Check with the manufacturer for specific repair recommendations and procedures.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## 87. Restraint System (Air Bag)

Source: [https://help.cccis.com/webhelp/motor/gte/Content/PassiveRestraintSystems\\_AirBagType\\_.htm](https://help.cccis.com/webhelp/motor/gte/Content/PassiveRestraintSystems_AirBagType_.htm)

### Restraint System (Air Bag)

NOTE: RESTRAINT SYSTEMS, REPLACEMENT COMPONENTS and INSPECTION COMPONENTS are FOR ESTIMATING PURPOSES ONLY. Wiring repairs should only be performed in accordance with vehicle manufacturer specifications. Air bag components should never be refinished. Refinishing these components will alter the original design and change deployment characteristics. Always refer to vehicle manufacturer's recommended repair procedures when servicing any air bag system.

Before repairing any air restraint system, disconnect and isolate the battery cable so that any back-up power supply is discharged for the time period stipulated by the manufacturer to prevent accidental deployment when working on the vehicle. All system components and mounting areas must be inspected before

energizing the system.

On some vehicles the windshield is considered an integral component of the overall system and has specific requirements regarding materials and procedures used when replacing a windshield. Factory recommended procedures must be followed when servicing this type of system. Some vehicles are equipped with dual stage air bags. The air bags are deployed at different rates depending on the severity of the crash, seat belt usage and driver position. If a dual stage air bag has deployed, it is possible that only one stage has deployed, not both stages. If both stages have not deployed, there is a danger of the second stage deploying if the air bag is not properly handled. On vehicles that are equipped with dual stage air bags, ensure the air bag system is disabled before performing any repair procedures. The air bag should then be properly disposed of. Refer to the vehicle manufacturer's recommended procedures for air bag disposal.

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## **88. Pickup Truck - Box Assembly**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/PickupTruck\\_BoxAssembly.htm](https://help.cccis.com/webhelp/motor/gte/Content/PickupTruck_BoxAssembly.htm)

### **Pickup Truck - Box Assembly**

#### **SPECIAL NOTATION:**

Some replacement components may or may not be supplied with duplicated OEM caulk / seam sealer. This is not included in R&R time and requires an on-the-spot evaluation.

Be sure to itemize both outside and inside panel refinish times when applicable.

Box R&R

Bumper assembly R&I

Fuel door

Lamps

License plate bracket

Tailgate R&I

Wheelhouse liner (plastic)

Wiring R&I

Accessory items

Body side moldings

Emblems & nameplates

Fuel tank

Mud flaps

Stripe tape, decals or overlays

Things to Consider

Add If Required

Adhesive Type Moldings, Nameplates & Emblems

Bolt Or Clip-On Moldings

Component Classification

Damaged or Non-Reusable Moldings and Fasteners

Decals, Stripe Tape & Overlays

Drilling Operations

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

Panel Bonding (Metal Adhesives)

R & I (Remove & Reinstall)

R & R (Remove & Replace)

Special Substrate Metals

Refinish Starting Point

Basic Color Coat Application

Parts Supplied In Color

Refinish Clear Coat Application

Refinish Time Listings

Refinish Time Premise

Refinish Processes

Clear Coat Finishes (Base Coat / Clear Coat)

Color Blend (Adjacent Panels)

Deductions to Basic Refinish Times (Refinish Overlap)

Panel and/or Component Designation

Two-Tone Refinishing (Second Color Tone Application)

Things to Know

Bagging (Cover Entire Vehicle Exterior)

Clear Coat Undamaged Panel

De-Nib & Polish

Edges Of New Parts (Edging)

Stone Chip Guard (Protective Material)

Three-Stage Color Blend (Adjacent Panels)

Three-Stage Finishes (Base / Mica / Clear Coat)

Wet / Dry Sand, Rub-Out & Buff

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## **89. Pickup Truck - Box Fenders**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/PickupTruck\\_BoxFenders.htm](https://help.cccis.com/webhelp/motor/gte/Content/PickupTruck_BoxFenders.htm)

### **Pickup Truck - Box Fenders**

Fender R&R

Fuel door

Lamp

License plate bracket

Wheelhouse liner (plastic)

Body side moldings

Emblems & nameplates

Fuel tank

Stripe tape, decals or overlays

Things to Consider

Add If Required

Adhesive Type Moldings, Nameplates & Emblems

Bolt Or Clip-On Moldings

Component Classification

Damaged or Non-Reusable Moldings and Fasteners

Decals, Stripe Tape & Overlays

Drilling Operations

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

Panel Bonding (Metal Adhesives)

R & I (Remove & Reinstall)

Special Substrate Metals

Refinish Starting Point

Basic Color Coat Application

Parts Supplied In Color  
Refinish Clear Coat Application  
Refinish Time Listings  
Refinish Time Premise  
Refinish Processes  
Clear Coat Finishes (Base Coat / Clear Coat)  
Color Blend (Adjacent Panels)  
Deductions to Basic Refinish Times (Refinish Overlap)  
Panel and/or Component Designation  
Two-Tone Refinishing (Second Color Tone Application)  
Things to Know  
Bagging (Cover Entire Vehicle Exterior)  
Clear Coat Undamaged Panel  
De-Nib & Polish  
Edges Of New Parts (Edging)  
Stone Chip Guard (Protective Material)  
Three-Stage Color Blend (Adjacent Panels)  
Three-Stage Finishes (Base / Mica / Clear Coat)  
Wet / Dry Sand, Rub-Out & Buff

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## **90. Pickup Truck - Box Front Panel**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/PickupTruck\\_BoxFrontPanel.htm](https://help.cccis.com/webhelp/motor/gte/Content/PickupTruck_BoxFrontPanel.htm)

### **Pickup Truck - Box Front Panel**

Adhesive application if required (all types)  
Bolt on extensions & fillers  
Caulk / seam sealer  
Cutting & welding as necessary  
Grind, fill & smooth welded seams (up to 150 grit sandpaper)  
Rivet removal and installation if required  
Body side moldings

Box R&I

Box set back

Standard running boards & step plates

Stripe tape, decals or overlays

Things to Consider

Add If Required

Bolt Or Clip-On Moldings

Component Classification

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

Panel Bonding (Metal Adhesives)

R & I (Remove & Reinstall)

Special Substrate Metals

Refinish Starting Point

Basic Color Coat Application

Parts Supplied In Color

Refinish Clear Coat Application

Refinish Time Listings

Refinish Time Premise

Refinish Processes

Clear Coat Finishes (Base Coat / Clear Coat)

Color Blend (Adjacent Panels)

Deductions to Basic Refinish Times (Refinish Overlap)

Panel and/or Component Designation

Things to Know

Bagging (Cover Entire Vehicle Exterior)

Clear Coat Undamaged Panel

De-Nib & Polish

Edges Of New Parts (Edging)

Three-Stage Color Blend (Adjacent Panels)

Three-Stage Finishes (Base / Mica / Clear Coat)

Weld Zone / Adjacent Panel

Wet / Dry Sand, Rub-Out & Buff

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## **91. Pickup Truck - Box Set Back**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/PickupTruck\\_BoxSetBack.htm](https://help.cccis.com/webhelp/motor/gte/Content/PickupTruck_BoxSetBack.htm)

### **Pickup Truck - Box Set Back**

#### **SPECIAL NOTATION:**

Pick-up box set back is considered by MOTOR to be the removal of the fasteners securing the box to the frame of the vehicle, D&R necessary wiring, D&R fuel neck, and moving the box rearward. This operation is generally used for accessing the cab back or side panels or welds on the outer panel. On newer model vehicles this operation may not be applicable due to space limitations for the pickup box to move back and BOX ASSEMBLY R&I would be used.

Disconnect and move box rearward

Move box forward and reconnect

Fuel tank

Bumper R&I

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## **92. Pickup Truck - Box Side Panels**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/PickupTruck\\_BoxSidePanels.htm](https://help.cccis.com/webhelp/motor/gte/Content/PickupTruck_BoxSidePanels.htm)

### **Pickup Truck - Box Side Panels**

#### **SPECIAL NOTATION:**

Some bolted-on Box Side Panel replacement components may or may not be supplied with duplicated OEM caulk/seam sealer. This is not included in R&R time and requires an on-the-spot evaluation.

Adhesive application if required (all types)

Bolt on extensions & fillers

Bumper assembly R&I

Caulk / seam sealer

Cutting & welding as necessary

Fuel door

Grind, fill & smooth welded  
seams (up to 150 grit sandpaper)  
Lamp  
Rivet removal and installation if required  
Tail gate R&I  
Wheelhouse liner (plastic)  
Body side moldings  
Box R&I  
Box set back  
Emblems & nameplates  
Front panel  
Fuel tank  
Outer fender  
Pickup box floor  
Standard running boards & step plates  
Stripe tape, decals or overlays  
Wheelhouse  
Things to Consider  
Add If Required  
Adhesive Type Moldings, Nameplates & Emblems  
Bolt Or Clip-On Moldings  
Component Classification  
Damaged or Non-Reusable Moldings and Fasteners  
Decals, Stripe Tape & Overlays  
Drilling Operations  
Footnotes  
Included and/or Not Included Labor Operations  
Labor Time Listings  
Labor Time Premise  
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R & I (Remove & Reinstall)  
Special Substrate Metals  
Refinish Starting Point  
Basic Color Coat Application  
Parts Supplied In Color  
Refinish Clear Coat Application  
Refinish Time Listings

Refinish Time Premise

Refinish Processes

Clear Coat Finishes (Base Coat / Clear Coat)

Color Blend (Adjacent Panels)

Deductions to Basic Refinish Times (Refinish Overlap)

Panel and/or Component Designation

Two-Tone Refinishing (Second Color Tone Application)

Things to Know

Bagging (Cover Entire Vehicle Exterior)

Clear Coat Undamaged Panel

De-Nib & Polish

Edges Of New Parts (Edging)

Stone Chip Guard (Protective Material)

Three-Stage Color Blend (Adjacent Panels)

Three-Stage Finishes (Base / Mica / Clear Coat)

Weld Zone / Adjacent Panel

Wet / Dry Sand, Rub-Out & Buff

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## **93. Pickup Truck - Box Tail Gate**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/PickupTruck\\_BoxTailGate.htm](https://help.cccis.com/webhelp/motor/gte/Content/PickupTruck_BoxTailGate.htm)

### **Pickup Truck - Box Tail Gate**

#### **SPECIAL NOTATION:**

Some replacement components may or may not be supplied with duplicated OEM caulk / seam sealer. This is not included in R&R time and requires an on-the-spot evaluation.

Disconnect at hinge.

All bolted on parts

Hinge (bolted type)

Wiring

Align to vehicle

Body side moldings

Emblems & nameplates  
Hinge (welded type)  
Stripe tape, decals or overlays  
Things to Consider  
Add If Required  
Adhesive Type Moldings, Nameplates & Emblems  
Bolt Or Clip-On Moldings  
Component Classification  
Damaged or Non-Reusable Moldings and Fasteners  
Decals, Stripe Tape & Overlays  
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Color Blend (Adjacent Panels)  
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Panel and/or Component Designation  
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Bagging (Cover Entire Vehicle Exterior)  
Clear Coat Undamaged Panel  
De-Nib & Polish  
Edges Of New Parts (Edging)  
Three-Stage Color Blend (Adjacent Panels)  
Three-Stage Finishes (Base / Mica / Clear Coat)  
Wet / Dry Sand, Rub-Out & Buff

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## **94. Pickup Truck - Cab Assembly Skeleton**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/PickupTruck\\_CabAssembly\\_Skeleton.htm](https://help.cccis.com/webhelp/motor/gte/Content/PickupTruck_CabAssembly_Skeleton.htm)

### **Pickup Truck - Cab Assembly Skeleton**

Bonded glass installation

Cab R&R

Carpeting

Doors R&I

Electrical wiring in cab R&I

Glass (fixed / moveable)

Headliner

Instrument panel R&I

Interior trim

Interior mirror

Reveal moldings

Seats R&I

Steering column R&I

Body side molding

Bonded glass removal from damaged cab

Brackets & braces transfer

Caulk / seam sealer

Electrical Wiring

Emblems & nameplates

Exterior lamps

Foam

Front end sheet metal

Lock cylinder recoding

Pickup box set back

Stripe tape, decals or overlays

Sun roof

Things to Consider

Acoustical and Structural Foam Fillers

Add If Required

Drilling Operations

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

Panel Bonding (Metal Adhesives)

R & I (Remove & Reinstall)

R & R (Remove & Replace)

Sectioning

Special Substrate Metals

Structural Component Identification

Structural Glass

Refinish Starting Point

Basic Color Coat Application

Refinish Clear Coat Application

Refinish Time Listings

Refinish Processes

Clear Coat Finishes (Base Coat / Clear Coat)

Color Blend (Adjacent Panels)

Deductions to Basic Refinish Times (Refinish Overlap)

Panel and/or Component Designation

Things to Know

Bagging (Cover Entire Vehicle Exterior)

Clear Coat Undamaged Panel

De-Nib & Polish

Edges Of New Parts (Edging)

Three-Stage Color Blend (Adjacent Panels)

Three-Stage Finishes (Base / Mica / Clear Coat)

Weld Zone / Adjacent Panel

Wet / Dry Sand, Rub-Out & Buff

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## **95. Pickup Truck - Cab Side / Corner / Back Panel**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/PickupTruck\\_CabSide\\_Corner\\_BackPanel.htm](https://help.cccis.com/webhelp/motor/gte/Content/PickupTruck_CabSide_Corner_BackPanel.htm)

### **Pickup Truck - Cab Side / Corner / Back Panel**

Adhesive application if required (all types)

Carpet & insulation turn back

Caulk / seam sealer

Cutting & welding as necessary

Grind, fill & smooth welded seams (up to 150 grit sandpaper)

Rear seat R&I (extended or crew cabs)

Rivet removal and installation if required

Seat R&I (regular cab)

Side glass & moldings

Sill plate

Seatbelt anchor

Weatherstrips

Back glass

Body side moldings

Box set back

Carpet & insulation R&I

Emblems & nameplates

Exterior lamp

Foam

Headliner

Interior trim

Stripe tape, decals or overlays

Things to Consider

Acoustical and Structural Foam Fillers

Add If Required

Adhesive Type Moldings, Nameplates & Emblems

Damaged or Non-Reusable Moldings and Fasteners

Decals, Stripe Tape & Overlays

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings  
Labor Time Premise  
Panel Bonding (Metal Adhesives)  
R & I (Remove & Reinstall)  
Sectioning  
Special Substrate Metals  
Structural Component Identification  
Structural Glass  
Refinish Starting Point  
Basic Color Coat Application  
Parts Supplied In Color  
Refinish Clear Coat Application  
Refinish Time Listings  
Refinish Time Premise  
Refinish Processes  
Clear Coat Finishes (Base Coat / Clear Coat)  
Color Blend (Adjacent Panels)  
Deductions to Basic Refinish Times (Refinish Overlap)  
Panel and/or Component Designation  
Two-Tone Refinishing (Second Color Tone Application)  
Things to Know  
Bagging (Cover Entire Vehicle Exterior)  
Clear Coat Undamaged Panel  
De-Nib & Polish  
Edges Of New Parts (Edging)  
Stone Chip Guard (Protective Material)  
Three-Stage Color Blend (Adjacent Panels)  
Three-Stage Finishes (Base / Mica / Clear Coat)  
Weld Zone / Adjacent Panel  
Wet / Dry Sand, Rub-Out & Buff

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## **96. Pickup Truck - Uniside without Side or Corner Panel**

## **Pickup Truck - Uniside without Side or Corner Panel**

### **SPECIAL NOTATION:**

When replacing welded door hinges, add one half of the Door Hinge R&R time to Uniside R&R time.

Adhesive application (all types)

Back panel trim

Carpet & insulation turn back

Caulk / seam sealer

Center pillar trim (4 door models)

Cutting and welding as necessary

Door striker

Front seat R&I

Grind, fill & smooth welded seams (up to 150 grit sandpaper)

Hinge pillar trim

Jamb switches

Rear seat R&I

Rivet removal and installation if required

Roof drip rail moldings

Seat belt anchor

Side panel trim

Sill plates

Weatherstrips / windlace

Back glass and moldings

Body side moldings

Box set back

Carpet & insulation R&I

Control cables

Doors

Door hinge

Electrical wiring

Emblems & nameplates

Fender

Fluid lines

Foam

Instrument panel

Road wheel

Side glass & moldings

Stripe tape, decals or overlay

Windshield glass and moldings

Things to Consider

Acoustical and Structural Foam Fillers

Add If Required

Adhesive Type Moldings, Nameplates & Emblems

Damaged or Non-Reusable Moldings and Fasteners

Decals, Stripe Tape & Overlays

Footnotes

Included and/or Not Included Labor Operations

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Labor Time Premise

Panel Bonding (Metal Adhesives)

R & I (Remove & Reinstall)

R & R (Remove & Replace)

Sectioning

Special Substrate Metals

Structural Glass

Refinish Starting Point

Basic Color Coat Application

Parts Supplied In Color

Refinish Clear Coat Application

Refinish Time Listings

Refinish Time Premise

Refinish Processes

Clear Coat Finishes (Base Coat / Clear Coat)

Color Blend (Adjacent Panels)

Deductions to Basic Refinish Times (Refinish Overlap)

Panel and/or Component Designation

Two-Tone Refinishing (Second Color Tone Application)

Things to Know

Bagging (Cover Entire Vehicle Exterior)

Clear Coat Undamaged Panel

De-Nib & Polish

Edges Of New Parts (Edging)

Stone Chip Guard (Protective Material)

Three-Stage Color Blend (Adjacent Panels)

Three-Stage Finishes (Base / Mica / Clear Coat)

Weld Zone / Adjacent Panel

Wet / Dry Sand, Rub-Out & Buff

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## **97. Quarter Panel Without Convertible Roof**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/QUARTER\\_PANEL\\_WITHOUT\\_CONVERTIBLE\\_ROOF.htm](https://help.cccis.com/webhelp/motor/gte/Content/QUARTER_PANEL_WITHOUT_CONVERTIBLE_ROOF.htm)

### **Quarter Panel Without Convertible Roof**

SPECIAL NOTATION:

Quarter panel R&R times are generally based upon replacing at the factory seams.

On some vehicles, it is not practical to replace at the factory roof seam.

If a sectioning operation is not listed, then the R&R time on the parts line represents the time to splice at the best area. GTE page Included: Quarter panel sectioning.

Adhesive application if required (all types)

Back glass & moldings

Bolt on extensions & fillers

Carpet & mats turn back

Caulk / seam sealer

Cutting & welding as necessary

Door striker

Fuel door

Grind, fill & smooth welded seams (up to 150 grit sandpaper)

Interior molding

Quarter glass & reveal moldings (unless noted)

Quarter interior trim

Rear seat R&I

Rear lamp assemblies R&I (standard equipment)

Rear bumper assembly and/or cover R&I (if necessary)

Rear package tray trim  
Rivet removal and installation if required  
Sill plates  
Speakers  
Trunk compartment trim  
Weatherstrips attached to quarter panel  
Wheelhouse liner (plastic)  
Air bag components  
Antenna  
Body side moldings  
Electrical wiring  
Emblems & nameplates  
Foam  
Fuel tank  
Glass regulator assembly  
Headliner R&I  
Lamp R&I (optional equipment)  
Ornamentation  
Mud guard  
Panels inward of quarter  
Road wheel  
Roof cover R&I or roll back  
Spoilers & flares (optional equipment)  
Stripe tape, decals or overlays  
Trunk lid, lift gate or tail gate R&I  
Things to Consider  
Acoustical and Structural Foam Fillers  
Add If Required  
Adhesive Type Moldings, Nameplates & Emblems  
Damaged or Non-Reusable Moldings and Fasteners  
Decals, Stripe Tape & Overlays  
Footnotes  
Included and/or Not Included Labor Operations  
Labor Time Listings  
Labor Time Premise  
Panel Bonding (Metal Adhesives)  
R & I (Remove & Reinstall)

Sectioning

Structural Component Identification

Structural Glass

Refinish Starting Point

Basic Color Coat Application

Parts Supplied In Color

Refinish Clear Coat Application

Refinish Time Listings

Refinish Time Premise

Refinish Processes

Clear Coat Finishes (Base Coat / Clear Coat)

Color Blend (Adjacent Panels)

Deductions to Basic Refinish Times (Refinish Overlap)

Panel and/or Component Designation

Two-Tone Refinishing (Second Color Tone Application)

Things to Know

Bagging (Cover Entire Vehicle Exterior)

Clear Coat Undamaged Panel

De-Nib & Polish

Edges Of New Parts (Edging)

Stone Chip Guard (Protective Material)

Three-Stage Color Blend (Adjacent Panels)

Three-Stage Finishes (Base / Mica / Clear Coat)

Weld Zone / Adjacent Panel

Wet / Dry Sand, Rub-Out & Buff

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## **98. Quarter Panel - Convertible Roof Models**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/QuarterPanel\\_ConvertibleRoofModels.htm](https://help.cccis.com/webhelp/motor/gte/Content/QuarterPanel_ConvertibleRoofModels.htm)

### **Quarter Panel - Convertible Roof Models**

Adhesive application if required (all types)

Belt moldings  
Bolt on extensions & fillers  
Carpet & insulation turn back  
Caulk / seam sealer  
Cutting & welding as necessary  
Door striker  
Fuel door  
Grind, fill & smooth welded  
seams (up to 150 grit sandpaper)  
Lamp (std equip)  
Quarter glass & regulator assembly  
Quarter interior trim  
Rear lamp assemblies R&I  
Rear seat R&I (if necessary)  
Rear bumper assembly and/or cover R&I (if necessary)  
Rivet removal and installation if required  
Sill plates  
Speakers  
Trunk compartment trim  
Weatherstrips attached to quarter panel, seal strip  
Wheelhouse liner (plastic)  
Air bag components  
Antenna  
Body side moldings  
Boot well  
Carpet & insulation R&I  
Convertible top alignment  
Electrical Wiring  
Emblems & nameplates  
Foam  
Fuel tank  
Mud guard  
Panels inward of quarter  
Power cylinder  
Retractable roof alignment  
Road wheel  
Roll bar

Spoilers & flares

Stripe tape, decals or overlays

Trunk lid R&I

Things to Consider

Structural Component Identification

Sectioning

R & I (Remove & Reinstall)

Panel Bonding (Metal Adhesives)

Labor Time Premise

Labor Time Listings

Included and/or Not Included Labor Operations

Footnotes

Decals, Stripe Tape & Overlays

Damaged or Non-Reusable Moldings and Fasteners

Adhesive Type Moldings, Nameplates & Emblems

Add If Required

Acoustical and Structural Foam Fillers

Refinish Starting Point

Basic Color Coat Application

Parts Supplied In Color

Refinish Clear Coat Application

Refinish Time Listings

Refinish Time Premise

Refinish Processes

Clear Coat Finishes (Base Coat / Clear Coat)

Color Blend (Adjacent Panels)

Deductions to Basic Refinish Times (Refinish Overlap)

Panel and/or Component Designation

Two-Tone Refinishing (Second Color Tone Application)

Things to Know

Bagging (Cover Entire Vehicle Exterior)

Clear Coat Undamaged Panel

De-Nib & Polish

Edges Of New Parts (Edging)

Stone Chip Guard (Protective Material)

Three-Stage Color Blend (Adjacent Panels)

Three-Stage Finishes (Base / Mica / Clear Coat)

Weld Zone / Adjacent Panel

Wet / Dry Sand, Rub-Out & Buff

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **99. Refinish Clear Coat Application**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/REFINISHCLEARCOATAPPLICATION.htm>

### **Refinish Clear Coat Application**

Most major paint manufacturers recommend that when performing refinish repairs on an OEM base coat/clear coat or multistage finishes, the application of clear coat must be extended to the nearest panel edge or breakpoint to qualify for their lifetime refinish warranties

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **100. Refinish Time Listings**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/REFINISHTIMELISTINGS.htm>

### **Refinish Time Listings**

All refinish times are listed in hours and tenths of an hour. A time in parentheses adjacent to the part name, such as (p3.5) indicates three and one half hours. Replacement operation time does not include time necessary to refinish the component.

Operation times for the application of painted on stripes are not covered in this publication. The time necessary to perform this type of operation should be estimated after an on-the-spot evaluation of required procedure.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **101. Refinish Time Premise**

## **Refinish Time Premise**

Published refinish times are for one color applied to new undamaged OEM replacement components, without exterior trim, interior trim or other attached components, and applied in one continuous process. For damaged panel(s), published refinish times may be applied after the damaged panel has been returned to a NEW UNDAMAGED condition.

Refinish times do not include time which may be required to match color tints or defective finish textures on interior or exterior surfaces. Refinish times do not include disassembly of components such as Mirrors, Grilles and Headlamps that may come serviced as an assembly. Nor do they include time which may be required to correct finish imperfections caused by improper weather conditions, application, or environmental contamination such as dust, dirt, grease, etc. Retrieval of accurate color information beyond paint chip and information available from the vehicle would not be included and would require an on-the-spot evaluation. MOTOR advises all parties consider these factors beforehand to determine mutually acceptable provisions in the event such conditions exist or occur.

### **ADAS PRECAUTIONS**

#### **ANTI-THEFT LABELS (R DOT)**

Replacement part labels are coded with the letter "R" to show that it is a replacement part. R Dot labels should not be removed from the part. Use caution when refinishing, rustproofing, or undercoating replacement components to avoid damaging the label.

#### **BUMPER COVERS AND OTHER FLEXIBLE COMPONENTS**

Refinish times listed on the parts detail lines for these components are based on the items being refinished prior to installation. Refinish time listed on the parts detail line for an OEM bumper cover that has both body color and un-painted grained portion allows for the refinish of the body color only. Masking the grained, textured, or non-body color portions in preparation for body color application is an included operation. It includes exterior surface and edges refinished during one continuous process. If a separate edging procedure is utilized then the appropriate time should be estimated after an on-the-spot evaluation. Refinish times do not include removal of mold release agent from new unprimed molded components. Parts received from the OEM manufacturer without primer and some non-OEM parts with or without primer should be tested for the presence of release agents that would cause paint adhesion problems and treated accordingly. An operation for the preparation of unprimed bumpers will be provided whenever such manufacturing information has been made available by the OEM. However, this information may not be available at the time of publication, and preparation for unprimed bumpers may still be required, even though a published operation is not present. Preparation time should be estimated after an on-the-spot evaluation. For unprimed component preparation time, see "Add if Required" operation(s). Unprimed Flexible Component Preparation .

#### **BUMPER REINFORCEMENTS AND IMPACT BARS**

MOTOR has determined that paint found on factory-installed impact bars and bumper reinforcements is an artifact of the factory refinish process (over-spray). Impact bars and bumper reinforcements may come from the factory refinished in various stages of completeness and quality. However, replicating the OE finish on factory-installed impact bars and bumper reinforcements may not satisfy the paint manufacturer's warranty requirements. See SPECIAL PRECATIONS, REFINISH CLEAR COAT APPLICATION for further information. These factors should be considered when developing the estimate after an on-the-spot evaluation.

#### **DOOR OUTER REPAIR PANELS**

Refinish times listed on the parts detail line for new repair panels (i.e. door outer repair panel, tailgate and liftgate repair panels) include panel lip. It does not include time for refinishing the entire door frame edge or interior side. Where possible, MOTOR will publish time for those areas under a "Refinishing Notes" heading within that group.

#### DOOR SHELLS, LIFTGATES, AND TAILGATES

Refinish times listed on the parts detail line for these new components include exterior surface, edges and interior sides, unless otherwise noted in text.

Refinish times listed under the "Refinishing Notes" heading for "door outer panel only" operations do not include time for refinishing the door frame edge or interior side. Where possible, MOTOR will publish time for those areas under a "Refinishing Notes" heading within that group.

#### FENDERS, HOODS, TRUNK LIDS AND OTHER MAJOR BOLTED PANELS

Refinish times listed on the parts detail line for these new panels do not include time for refinishing the edge or underside. Where possible, MOTOR will publish time for those areas under a "Refinishing Notes" heading within that group.

#### QUARTER PANELS AND OTHER MAJOR WELDED PANELS

Refinish times listed on the parts detail line for these new panels include exterior side, recessed edges, gutters and pockets, unless otherwise noted in text.

Refinish times listed under the "Refinishing Notes" heading for quarter panels or other major welded panels "exterior surface only" operations do not include time for refinishing recessed edges, gutters and pockets. Where possible, MOTOR will publish time for these areas under the "Refinishing Notes" heading within that group.

#### NEW UNDAMAGED PANEL

A component manufactured to the same exacting standards as the parts installed on new vehicles when the car was built. Exterior body panels are supplied with a smooth painted surface (e-coat).

#### UNDERSIDE COLORS

Refinish times presented in this guide are developed under the premise that the underside and jamb color is the same as the exterior body color. Some vehicle manufacturers use a different color for the engine compartment, trunk compartment and/or jambs. An additional paint mix is required if the underside and/or jamb color is a different color than the exterior body color. Clear coat (gloss or matte) will be required for base color coat applications. Some component backsides of vehicles may not be refinished; however, the OE replacement component may be supplied in a different E-coat color than the damaged vehicle. An on-the-spot evaluation should be performed to determine if the non-refinished panels require additional refinishing to match pre-accident condition. Replicating the OE finish may not satisfy the paint manufacturer's warranty requirements. See SPECIAL PRECAUTIONS, REFINISH CLEAR COAT APPLICATION for further information. These factors should be considered when developing the estimate after an on-the-spot evaluation.

#### PRIME & BLOCK

Prime & block (high build / primer-filler) is a required procedure that restores a repaired panel surface, including the joined areas of replaced welded panels, from 150-grit finish to that of a NEW UNDAMAGED condition. It is MOTOR's position that prime & block is a refinishing process best reserved for the judgment of an estimator / appraiser following a thorough on-the-spot evaluation of the specific vehicle and damage in question.

#### REPAIRED PANEL REFINISHING

MOTOR suggests using component(s) base refinish time for this type of procedure after the damaged panel is repaired to new undamaged condition. Repaired surface preparation requires an on the spot evaluation for

additional procedural steps such as featheredge and/or prime and block not required for new undamaged panels.

#### TRANSLUCENT COLORS

Some colors may need additional applications of material above the normal application of three coats due to pigments being translucent by nature. Any additional application of material for coverage should be considered when developing the estimate and may require an on-the-spot evaluation.

#### PARTIAL PANEL REFINISHING

MOTOR defines partial panel refinishing as refinishing an area within a damaged body panel or underneath body cladding after the panel has been repaired to that of a "NEW UNDAMAGED PANEL," for the purpose of creating an undetectable color transition.

It is MOTOR's position that partial panel refinishing is a process best reserved for the judgment of an estimator/appraiser following a thorough on-the-spot evaluation of the specific vehicle and refinish requirements in question. Refer to G.T.E. "BASIC COLOR COAT APPLICATION."

#### MATTE/FLAT REFINISHES

MOTOR refinish times and clear coat formula is applicable for use with Matte/Flat finishes when the following Not Included items are factored in the refinish time: an extra clean environment needs to be maintained to reduce air-borne contamination; performing a test spray-out or let down panel to match gloss or lack of gloss; tinting the matte clear to adjust the level of gloss from a paint additive to reduce the gloss level; application of a standard mix clear coat and then sanding the clear coat so the matte finish matches the same texture of the adjacent panels; any additional masking of adjacent panels. These additional steps should be considered when developing the estimate.

#### MATTE AND GLOSS CLEAR REFINISHES COMBINED

Some vehicles may have two different clear coat finishes applied and would require a separate spray booth cycle in order to properly refinish the vehicle. In instances where this is necessary this should be treated as a separate refinish operation.

#### 4 STAGE REFINISH

In addition to the base and mid coat applications of a 3-Stage process, the 4-Stage refinish process, depending on paint manufacturer, may possibly require the application of an additional ground coat. Some paint manufacturers may use a tinted clear coat in addition to the final non-tinted clear coat. These additional steps are not included and should be considered when developing an estimate using an on-the-spot evaluation. MOTOR does not offer a formula for 4-Stage refinish.

#### SPECIAL NOTATION:

The items or operations below were not considered during the development of any published basic refinish operation times. These operations may, or may not be required depending upon the vehicle or process used. If any of these items or operations are required, they should be considered by the estimator and added to the estimate if necessary.

### REFINISH, WET / DRY SAND, DE-NIB, and/or RUB-OUT TIME DOES NOT INCLUDE:

- Anti-corrosion material application
- Assembling/setting up a unique fixture/stand to hold parts for refinishing
- Filling, blocking, featheredging repaired panels
- Flex additive mixing time
- Flex prep application

- Mask inner panels ex: apron / cowl / pillars / rail / floor, etc
- Material /consumable product costs
- Molding & ornamentation
- Protective coating material application
- Protective coating removal
- Sound deadening application
- Spatter paint application time
- Stripe tape, decal & overlay
- Waste disposal fees (all types)

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **102. Rocker Panel**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/ROCKERPANEL.htm>

### **Rocker Panel**

Adhesive application if required (all types)

Carpet & insulation turn back

Caulk / seam sealer

Cutting & welding as necessary

Grind, fill & smooth welded seams (up to 150 grit sandpaper)

Interior trim panels

Rivet removal and installation if required

Sill plates

Weatherstrips pull back

Carpet, insulation or seat R&I

Control cables

Door

Emblems & nameplates

Fender

Foam

Fuel lines

Moldings

Stripe tape, decals

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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### **103. R & I (Remove & Reinstall)**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/R\\_I\\_Remove\\_Reinstall\\_.htm](https://help.cccis.com/webhelp/motor/gte/Content/R_I_Remove_Reinstall_.htm)

#### **R & I (Remove & Reinstall)**

Item is removed, set aside and later reinstalled and aligned for proper fit (does not include Suspension / Wheel alignment). Generally used to gain access to another part.

If a R&I time is not available, published R&R times can be used when the steps required to R&R a component are the same procedure steps required to R&I a component. Example; bolt or clip on moldings, ornamentation, mirrors, trim, door handles, locks, cylinders, latches, and many single piece components.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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### **104. R & R (Remove & Replace)**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/R\\_R\\_Remove\\_Replace\\_.htm](https://help.cccis.com/webhelp/motor/gte/Content/R_R_Remove_Replace_.htm)

#### **R & R (Remove & Replace)**

Remove old part, transfer necessary items to new part (unless otherwise noted in Labor Procedure pages), replace and align. Does not include Suspension/Wheel alignment.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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### **105. Rear Body - Floor Pan / Rails / Crossmembers**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/RearBody\\_FloorPan\\_Rails\\_Crossmembers.htm](https://help.cccis.com/webhelp/motor/gte/Content/RearBody_FloorPan_Rails_Crossmembers.htm)

## **Rear Body - Floor Pan / Rails / Crossmembers**

Adhesive application if required (all types)

Cutting & welding as necessary

Caulk / seam sealer

Rear body panel (on floor pan only)

Rivet removal and installation if required

Bolted on parts

Electrical wiring

Exhaust system

Rear body panel (on rails only)

Sound deadening pad

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **106. Rear Body - Lower Body Panel**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/RearBody\\_LowerBodyPanel.htm](https://help.cccis.com/webhelp/motor/gte/Content/RearBody_LowerBodyPanel.htm)

### **Rear Body - Lower Body Panel**

Adhesive application if required (all types)

Bolt on extensions & fillers

Caulk / seam sealer

Cutting & welding as necessary

Grind, fill & smooth welded seams (up to 150 grit sandpaper)

Lamp R&I

Latch or striker

License plate bracket

License plate

Rear compartment trim (as necessary)

Rivet removal and installation if required

Sill plate

Weatherstrips

Camera

Electrical wiring

Fuel tank R&I  
Emblems & nameplates  
Energy absorber  
Exhaust system  
Lock cylinder recoding  
Moldings  
Stripe tape, decals or overlays  
Valance panel

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **107. Rear Bumper Cover - R&R Cover Type**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/RearBumperAssemblyR\\_RFasciaType.htm](https://help.cccis.com/webhelp/motor/gte/Content/RearBumperAssemblyR_RFasciaType.htm)

### **Rear Bumper Cover - R&R Cover Type**

#### **SPECIAL NOTATION:**

The removal of mold-release agents from and preparation/priming of unprimed bumpers is required and is NOT INCLUDED. Where the OEM has provided information regarding the availability of unprimed bumper covers, an "Add if Required" labor operation has been added to the bumper group.

Align to vehicle  
Emblem & nameplate  
Energy absorber (when mounted to cover)  
Lamps (when mounted to bumper)  
License plate/bracket (if attached)  
Molding & impact strip  
Rear bumper cover  
Reflectors  
Tow eye cap  
Antenna  
Bumper brackets/frame rail extensions (if mounted to frame rail)  
Hands free motion/kick sensor (if mounted to bumper)  
Lamps (optional equipment, or not mounted to bumper)  
Reinforcement/impact bar  
Reverse sensor

Stripe tape, decals or overlays

Trunk trim / liner

Valance panel / spoiler

Things to Consider

Add If Required

Adhesive Type Moldings, Nameplates & Emblems

Damaged or Non-Reusable Moldings and Fasteners

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

Part Quantities

Refinish Starting Point

Basic Color Coat Application

Parts Supplied In Color

Refinish Clear Coat Application

Refinish Time Listings

Refinish Time Premise

Unprimed Flexible Component Preparation

Refinish Processes

Clear Coat Finishes (Base Coat / Clear Coat)

Panel and/or Component Designation

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **108. Rear Bumper Assembly - Overhaul Face Bar Type**

Source:

[https://help.cccis.com/webhelp/motor/gte/Content/RearBumperAssembly\\_OverhaulFaceBarType.htm](https://help.cccis.com/webhelp/motor/gte/Content/RearBumperAssembly_OverhaulFaceBarType.htm)

### **Rear Bumper Assembly - Overhaul Face Bar Type**

Align to vehicle

Disassemble complete assembly

Emblem & nameplate

Lamp  
Lamps (when mounted to bumper)  
License plate/bracket (if attached)  
Molding & impact strip  
Reassemble complete assembly  
Reflectors  
Replace damaged parts  
Energy absorber,if mounted to frame rail (all types)  
Lamps (optional equipment, or not mounted to bumper)  
Reverse sensor  
Stripe tape, decals or overlays  
Trunk trim / liner  
Valance panel / spoiler  
Things to Consider  
Add If Required  
Adhesive Type Moldings, Nameplates & Emblems  
Damaged or Non-Reusable Moldings and Fasteners  
Footnotes  
Included and/or Not Included Labor Operations  
Labor Time Listings  
Labor Time Premise  
Part Quantities  
Refinish Starting Point  
Basic Color Coat Application  
Parts Supplied In Color  
Refinish Clear Coat Application  
Refinish Time Listings  
Refinish Time Premise  
Refinish Processes  
Clear Coat Finishes (Base Coat / Clear Coat)  
Panel and/or Component Designation

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## 109. Rear Bumper Cover - Overhaul Cover Type

Source: [https://help.cccis.com/webhelp/motor/gte/Content/RearBumperAssembly\\_OverhaulFasciaType.htm](https://help.cccis.com/webhelp/motor/gte/Content/RearBumperAssembly_OverhaulFasciaType.htm)

### Rear Bumper Cover - Overhaul Cover Type

#### SPECIAL NOTATION:

The removal of mold-release agents from and preparation/priming of unprimed bumpers is required and is NOT INCLUDED. Where the OEM has provided information regarding the availability of unprimed bumper covers, an "Add if Required" labor operation has been added to the bumper group.

Align to vehicle

Disassemble cover assembly

Emblem & nameplate

Energy absorber (when mounted to bumper)

Lamps (when mounted to bumper)

License plate/bracket (if attached)

Molding & impact strip

Reassemble complete assembly

Reflectors

Replace damaged parts

Tow eye cap

Antenna

Bumper brackets/frame rail extensions (if mounted to frame rail)

Hands free motion/kick sensor (if mounted to bumper)

Lamps (optional equipment, or not mounted to bumper)

Reinforcement/impact bar

Reverse sensor

Stripe tape, decals or overlays

Trunk trim / liner

Valance panel / spoiler

Things to Consider

Adhesive Type Moldings, Nameplates & Emblems

Damaged or Non-Reusable Moldings and Fasteners

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

Overhaul

Part Quantities  
Refinish Starting Point  
Basic Color Coat Application  
Parts Supplied In Color  
Refinish Clear Coat Application  
Refinish Time Listings  
Refinish Time Premise  
Unprimed Flexible Component Preparation  
Refinish Processes  
Clear Coat Finishes (Base Coat / Clear Coat)  
Panel and/or Component Designation

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **110. Rear Bumper Assembly - R & I All Types**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/RearBumperAssembly\\_R\\_IAllTypes.htm](https://help.cccis.com/webhelp/motor/gte/Content/RearBumperAssembly_R_IAllTypes.htm)

### **Rear Bumper Assembly - R & I All Types**

Align to vehicle  
Emblems & nameplate  
Energy absorber, all types  
Molding & impact strip  
Stripe tape, decals or overlays  
Trunk trim / liner  
Things to Consider  
Damaged or Non-Reusable Moldings and Fasteners  
Included and/or Not Included Labor Operations  
Labor Time Listings  
Labor Time Premise  
R & I (Remove & Reinstall)

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **111. Rear Bumper Assembly - R & R Face Bar Type**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/RearBumperAssembly\\_R\\_RFaceBarType.htm](https://help.cccis.com/webhelp/motor/gte/Content/RearBumperAssembly_R_RFaceBarType.htm)

### **Rear Bumper Assembly - R & R Face Bar Type**

Align to vehicle

Emblem & nameplate

Face bar R&R

Lamps (when mounted to bumper)

License plate/bracket (if attached)

Molding & impact strip

Energy absorber,if mounted to frame rail (all types)

Lamps (optional equipment, or not mounted to bumper)

Reverse sensor

Stripe tape, decals or overlays

Trunk trim / liner

Valance panel / spoiler

Things to Consider

Adhesive Type Moldings, Nameplates & Emblems

Damaged or Non-Reusable Moldings and Fasteners

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

Overhaul

Part Quantities

Refinish Starting Point

Basic Color Coat Application

Parts Supplied In Color

Refinish Clear Coat Application

Refinish Time Listings

Refinish Time Premise

Refinish Processes

Clear Coat Finishes (Base Coat / Clear Coat)

Panel and/or Component Designation

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **112. Rear Lamps**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/RearLamps.htm>

### **Rear Lamps**

#### **SPECIAL NOTATION:**

On some vehicles, the trunk compartment trim is partially pulled back rather than completely removed in order to access the rear lamp fasteners. MOTOR has taken this operation into consideration, and the partial pull back of trunk compartment trim is included in Rear Lamp R&R when applicable.

Access cover

Bulb

Trunk compartment trim (partial pull back if necessary)

Trunk compartment trim

Front Lamps

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **113. Rear Suspension - Individual Operations**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/RearSuspension\\_IndividualOperations.htm](https://help.cccis.com/webhelp/motor/gte/Content/RearSuspension_IndividualOperations.htm)

### **Rear Suspension - Individual Operations**

Raise & support vehicle (1 time)

Bolted on parts

Brake calipers R&I

Brake shoes or pads

Brakes adjust & bleed

Road wheel

Suspension / wheel alignment

Wheel cylinders

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **114. Rear Suspension - Overhaul Independent Type**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/RearSuspension\\_OverhaulIndependentType.htm](https://help.cccis.com/webhelp/motor/gte/Content/RearSuspension_OverhaulIndependentType.htm)

### **Rear Suspension - Overhaul Independent Type**

ABS sensor

Ball joint

Bearing

Bearing seal R&R

Brake drum / rotor

Coil spring components (if equipped)

Hub

Knuckle

Lower control arm components

Overhaul procedure as defined

Radius rod & bushing D&R

Raise & support vehicle (1 time)

Shock absorber

Splash shield

Stabilizer link

Strut

Torsion bar (if equipped)

Upper control arm components

Axle or carrier assy

Brake pad or shoe

Brakes adjust & bleed

Crossmember

Road wheel

Stabilizer bar

Suspension / wheel alignment

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **115. Rear Suspension - Overhaul Rear Trailing Type**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/RearSuspension\\_OverhaulRearTrailingType.htm](https://help.cccis.com/webhelp/motor/gte/Content/RearSuspension_OverhaulRearTrailingType.htm)

### **Rear Suspension - Overhaul Rear Trailing Type**

ABS sensor

Axle stub / spindle

Axle beam

Axle mounting pads

Bearing

Bearing seals R&R

Brake rotor

Coil spring or leaf spring components

Control arms R&I

Overhaul procedure as defined

Radius rod & bushing

Raise & support vehicle (1 time)

Shock absorber

Splash shield

Stabilizer link

Strut

Axle or carrier assy

Brake pad or shoes

Brakes adjust & bleed

Drive axles

Road wheel

Stabilizer bar

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## 116. Rear Suspension - Overhaul Strut Type

Source: [https://help.cccis.com/webhelp/motor/gte/Content/RearSuspension\\_OverhaulStrutType.htm](https://help.cccis.com/webhelp/motor/gte/Content/RearSuspension_OverhaulStrutType.htm)

### Rear Suspension - Overhaul Strut Type

ABS sensor

Bearing seal R&R

Bearing

Brake drum / rotor

Coil spring components

Hub

Knuckle

Lower control arm

Lower control arm bushing

Overhaul procedure as defined

Radius rod & bushing

Raise & support vehicle (1 time)

Splash shield

Stabilizer link

Strut seat

Strut radius rod & bushing D&R

Strut cartridge / assembly

Axle or carrier assy

Bolt-on crossmember

Brake pad or shoes

Brakes adjust & bleed

Rear drive axle carrier / differential

Road wheel

Stabilizer bar

Suspension / wheel alignment

Wheel cylinders

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **117. Road Wheel /Tire - R & R**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/RoadWheel\\_R\\_R.htm](https://help.cccis.com/webhelp/motor/gte/Content/RoadWheel_R_R.htm)

### **Road Wheel /Tire - R & R**

#### **SPECIAL NOTATION:**

When required, an additional 0.3 hrs. may be necessary to remove a spare tire from its storage location and install it on the hub, later remove the spare tire from hub and return it to the original storage location. Generally used when a road wheel is removed for repair and the spare tire is temporarily installed in its place.

Raise & support vehicle

Remove & reinstall wheel / tire assembly

Transfer /R&R tire

Lower vehicle

Torque wheel to OE specifications

Valve stem

TPMS sensor, if attached to valve stem

Balancing

Cost of valve stems or weights

Lock type lug nuts

Lock type wheel covers

Scan tool clear / reset

Tire disposal fee

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **118. Roof Convertible Top R&R**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/Roof\\_ConvertibleTopR\\_R.htm](https://help.cccis.com/webhelp/motor/gte/Content/Roof_ConvertibleTopR_R.htm)

### **Roof Convertible Top R&R**

Boot well

Fabric cover R&R

Frame assembly R&R

Quarter trim panels

Weatherstrips & retainers

Deck lid

Frame assembly alignment

Moldings & ornamentation

Power cylinder

Roll bar

Windshield frame

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **119. Roof - Headliner**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/Roof\\_Headliner.htm](https://help.cccis.com/webhelp/motor/gte/Content/Roof_Headliner.htm)

### **Roof - Headliner**

SPECIAL NOTATION:

This section applies to passenger cars, trucks and vans

Bezels (not included in R&I)

Coat hooks

Grab handles

High-mounted stop lamp (if attached to headliner)

Lamp (dome, reading)

Pads/Insulation (not included in R&I)

Reveal & interior Upper moldings / trim

Sunvisor

Sunvisor bracket

Transfer Wiring (not included in R&I)

Weatherstrip D&R

Head air bags

Interior lower moldings/trim

Overhead console

Seats (front or rear)

Weatherstrip R&I

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **120. Roof - Roof Panel R&R**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/Roof\\_RoofPanelR\\_R\\_Automobile\\_.htm](https://help.cccis.com/webhelp/motor/gte/Content/Roof_RoofPanelR_R_Automobile_.htm)

### **Roof - Roof Panel R&R**

#### **SPECIAL NOTATION:**

This section applies to passenger cars, SUVs , trucks and vans. Panoramic roof glass does not include removal of old adhesive, clean and preparation of sealing surfaces (vehicle and/or glass) and should be estimated after an on-the- spot evaluation.

Adhesive application if required (all types)

Back window & reveal moldings

Caulk / seam sealer

Cutting & welding as necessary

Flutter material

Glass run channels

Grind, fill & smooth welded seams (up to 150 grit sandpaper)

Headliner R&I, coat hooks & grab handles

High-mounted stop lamp (if attached to roof)

Lamp (dome, reading)

Liftgate R&I

Moldings

OEM sunroof or "T" top R&I

Rear view mirror / support R&I (attached to roof)

Reveal & interior Upper moldings / trim

Rivet removal and installation if required

Sunvisor

Upper quarter trim panels

Weatherstrips & retainers

Windshield glass

Electrical wiring

Emblems & nameplates

GPS Antenna

Head air bags  
Header panels  
Interior lower moldings/trim  
Lamp (exterior)  
Lift gate hinges  
Luggage racks  
Overhead console  
Rear package tray trim  
Roof covers  
Roof rails (exterior trim)  
Stripe tape, decals

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **121. Sectioning**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/SECTIONING.htm>

### **Sectioning**

Sectioning is an industry accepted procedure which involves replacing a vehicle's welded body panel (quarter, rocker, rail etc.) onto the undamaged portion of a vehicle body at a location other than at a factory seam. Special care and training is required when sectioning any part due to the wide use of special steels and vehicle design. Repairs of this type should only be performed according to the OEM service procedure .

The sectioning location may be affected after an on-the-spot evaluation of the vehicle's inner structure design.

The estimated work time premise (included / not included operations) for a sectioned panel is considered to be the same as for the full panel replacement, unless otherwise specified in a footnote attached to the sectioning operation. The on-the-spot evaluation will define inclusions/non-inclusions unique to the collision scenario and/or the sectioning approach chosen.

Some sectioning operations are not subject to Overlap Operations listed in the Headnotes of varying component groups. Adjacent components factored into Overlap Operation considerations may not apply due to the sectioning approach being implemented. Overlap deductions should be determined after an on-the-spot evaluation once sectioning approach has been determined.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **122. Stone Chip Guard (Protective Material)**

Source:

[https://help.cccis.com/webhelp/motor/gte/Content/STONE\\_CHOP\\_GUARD\\_Protective\\_Material\\_.htm](https://help.cccis.com/webhelp/motor/gte/Content/STONE_CHOP_GUARD_Protective_Material_.htm)

### **Stone Chip Guard (Protective Material)**

Vehicle manufacturers may apply a spray-on, chip-resistant coating to protect the vehicle's finish from chipping. This type of coating may be named differently depending on manufacturer i.e.; Stone Guard, Soft-chip Primer, PVC Chipping Primer, Chip Guard, Gravel Guard etc. This type of coating is designed to reduce paint chipping; appearance varies from textured surface to a smooth surface. Chip-resistant coatings may be applied to the vehicle's lower body and/or leading edges of the body. Refer to OEM service repair information for specific location, repair recommendations and/or replacement product(s). MOTOR recommends following OEM service repair information for replacement of these materials following a thorough on-the-spot evaluation of vehicle and damage in question.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **123. Structural Component Identification**

Source:

[https://help.cccis.com/webhelp/motor/gte/Content/STRUCTURAL\\_COMPONENT\\_IDENTIFICATION\\_1.htm](https://help.cccis.com/webhelp/motor/gte/Content/STRUCTURAL_COMPONENT_IDENTIFICATION_1.htm)

### **Structural Component Identification**

Welded structural parts can be made from different types of metal; identification is not based on metal type. Replacement requires specific measuring equipment and vehicle dimensions must be correct. If improperly repaired, road performance and crashworthiness of the vehicle may be affected. Airbag deployment may also be affected.

Some parts that are bolted, bonded or welded may add structural integrity to a vehicles body. These parts are made from different types of materials and vehicle dimensions must be correct before replacement. If improperly repaired, road performance and crashworthiness of the vehicle may be affected. Airbag deployment may also be affected.

Due to variations in vehicle construction from one OE manufacturer to another, structural qualifications may differ, impacting how MOTOR may qualify a component. A review of the OE service information at the time of estimate preparation may be needed to determine an OE manufacturer component classification.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## 124. Special Substrate Metals

Source: <https://help.cccis.com/webhelp/motor/gte/Content/SpecialSteels.htm>

### Special Substrate Metals

Some vehicle manufacturers use special or alternative metals in the construction of their vehicles. When possible, parts made from one of the ten listed materials will be identified using the abbreviations shown below.

It is not always possible or practical to identify where special or alternative metals are used on a vehicle. Do not assume that a part not identified as special metal isn't actually made of special metal. When in doubt, it is recommended that you contact the vehicle manufacturer for further information. When performing repairs to special metal substrates, refer to OEM recommendations for specific information. Improper repairs may adversely affect the structural integrity and safety of the vehicle.

In instances where a component is comprised of multiple grades of material, MOTOR will indicate the highest-grade material used. Therefore, it is necessary to refer to the relevant OEM repair information to determine if component repair or replacement is required based on the precise area damaged.

#### ALUMINUM (ALU)

Aluminum is lighter than steel and is non-magnetic. Aluminum is sensitive to galvanic corrosion. Only use fasteners, tools and materials that are designated for use only on aluminum. Heat limitations **MUST** be observed when working with this type of metal. Variances can be found in alloy, strength, temper, and gauge properties. Such variations need to be considered prior to attempting a repair, and an on-the-spot evaluation is required. Aluminum component repair and/or replacement often requires specific equipment and tools. Always consult OEM repair procedures for vehicle-specific information.

#### BORON STEEL (BOR)

Boron steel has extremely high strength; do not use heat on this type of metal. It cannot be straightened because of the extremely high heat used in the forming process. The only solution for damage is replacement. Boron steel can be cut with a cutoff wheel. Drilling boron steel is very difficult and may be accomplished using a titanium drill bit combined with slow speed (490 rpm).

#### CARBON FIBER COMPOSITE (CFC)

Visible weave carbon fiber (non-painted) and non-visible weave carbon fiber (painted) parts are extremely lightweight and strong. Structural carbon fiber parts will require replacement if damaged. Manufacturers may have repair options for non-structural parts such as exterior panels, provided that the damage does not extend to an edge. Reparability options depend on the manufacturer's recommendations and procedures. Only use fasteners, tools and materials that are designated for use on carbon fiber.

#### High Strength Steel (HSS)

Generally, frame rails, inner rocker panels, suspension crossmembers, and upper apron rails are made of this material (this varies by manufacturer). Heat limitations **MUST** be observed when working with this type of steel. When in doubt, most manufacturers recommend all steel be treated as HSS steel.

#### HYDROFORMED STEEL (HYD)

Hydroformed parts may be sectioned depending on the vehicle manufacturer's recommendation. Heat limitations **MUST** be observed when working with this type of steel.

#### MAGNESIUM (MAG)

Magnesium is much lighter, stronger, and more resistant to corrosion than steel. Magnesium has the tendency to crack or break either from collision damage or during straightening due to its quick work-hardening characteristics. Magnesium is also not weldable with common collision repair facility GMA (MIG) welding capabilities. Do not use oxyacetylene equipment or plasma arc cutting equipment around this or any other magnesium part. If magnesium catches on fire, it requires a class D fire extinguisher

#### SANDWICHED STEEL (SAS)

This type of steel (OEM's may have other names) is being used on the dash panels, floor panels and oil pans etc. This product has an engineered non-steel layer sandwiched between two cold rolled layers of steel; do not use heat on this type of metal unless otherwise recommended by OEM. It is used to help lower sound levels and vibration. This product may or may not be weld-able with common collision repair facility GMA (MIG) welding.

#### SHEET MOLDING COMPOUND (SMC)

A high strength glass-reinforced thermoset molding material which is normally compression-molded and is used on some vehicle body outer panels and radiator supports. Manufacturers may have repair options for these types of non-structural parts. Reparability options depend on the manufacturer's recommendations and procedures. Only use materials that are designated for use on sheet molded compounds.

#### STAINLESS STEEL (SSA):

Stainless steels are steel alloys that are very well known for their corrosion resistance, which increases with additional chromium content. Corrosion resistance may also be enhanced by nickel and molybdenum additions. Cosmetic repair to panels and paintless dent repair techniques are an acceptable method for repairing cosmetic panel damage. OEM repair guidelines should be followed.

#### ULTRA HIGH STRENGTH STEEL (UHS)

Generally, door guard beams, bumper reinforcements and other special purpose areas contain this type of steel. These parts must not be repaired. They are to be replaced only. Contact the vehicle manufacturer for applicable locations. This information supersedes information published in previous Guide to Estimating pages.

NOTE: Due to space limitations, special substrate information is not available in MOTOR e-Tech Online .

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **125. Stationary Side Glass (Doors, Quarter Panel, or Pillar)**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/StationarySideGlass\\_DoorsOrQuarterPanel\\_.htm](https://help.cccis.com/webhelp/motor/gte/Content/StationarySideGlass_DoorsOrQuarterPanel_.htm)

### **Stationary Side Glass (Doors, Quarter Panel, or Pillar)**

#### SPECIAL NOTATION:

Glass, moldings, mounting studs or mounting flange may be damaged or broken during normal service procedures on some applications. MOTOR advises that these factors be considered before performing glass operations in the event unavoidable damage occurs. Time for the removal of old adhesive, clean and preparation of sealing surfaces (vehicle and/or glass) should be estimated after an on the spot evaluation.

Reveal & interior moldings / trim

Rubber channel (if installed)

Test for leaks

Urethane adhesive application

Clean up of broken glass

Cost of installation material or kit

Refinish glass pinchweld (epoxy primer)

Things to Consider

Damaged or Non-Reusable Moldings and Fasteners

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

NAGS - Glass Prices

Structural Glass

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **126. Steering Column**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/Steering\\_Column.htm](https://help.cccis.com/webhelp/motor/gte/Content/Steering_Column.htm)

### **Steering Column**

Many vehicle manufacturers use collapsible steering columns to absorb energy sustained from a collision impact. These columns should be inspected for proper length, binding and deformation among other specific considerations. Failure to do so may prevent proper operation of steering column and/or air bag deployment. MOTOR recommends following vehicle manufacturer's guidelines for inspection and replacement of these components.

#### **Steering Column - Overhaul Operations**

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **127. Steering Column - Overhaul Operations**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/Steeringcolumn\\_overhauloperations.htm](https://help.cccis.com/webhelp/motor/gte/Content/Steeringcolumn_overhauloperations.htm)

## **Steering Column - Overhaul Operations**

### **SPECIAL NOTATION:**

Air bag considerations are included in all steering column Overhaul operations unless "ADD IF REQUIRED" heading is listed in the text.

Column housing

Column lower shroud

Column internal parts

Column upper shroud

Cruise control parts attached to column

Ignition lock cylinder

Ignition switch

Levers

Lower coupling

Shift lever

Shift bowl

Steering shafts

Steering shaft bearings

Steering wheel

Tilt mechanism

Turn signal switch / lever

Wiring

Electrical wiring

Lock cylinder recoding

Refinish column components

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **128. Steering Linkage - Individual Operations**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/Steeringlinkage\\_individualoperations.htm](https://help.cccis.com/webhelp/motor/gte/Content/Steeringlinkage_individualoperations.htm)

## **Steering Linkage - Individual Operations**

Individual component R&R

Brakes adjust & bleed

Road wheel

Suspension / wheel alignment

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **129. Stone Chip Guard (Protective Material)**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/StoneChipGuard\\_ProtectiveMaterial\\_.htm](https://help.cccis.com/webhelp/motor/gte/Content/StoneChipGuard_ProtectiveMaterial_.htm)

### **Stone Chip Guard (Protective Material)**

#### **SPECIAL NOTATION:**

The following items or operations were not considered during the development of any published basic refinish operation times. If any of these items or operations are required, they should be considered by the estimator.

- First panel;
  - Add 0.5 per panel
- Each additional panel;
  - Add 0.3 per panel

Up to a 12" height

Texture matching

Masking of non-applicable area(s)

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **130. Structural Glass**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/StructuralGlass.htm>

### **Structural Glass**

Urethane bonded stationary glass such as windshields, side / quarter glass and back glass adds structural integrity to a vehicles body and may be considered a structural component on many vehicles. Therefore, it is important to use proper materials and procedures when installing this type of glass. I-CAR and some vehicle manufacturers recommend the use of epoxy primer on glass pinchweld where coating has been removed. Utilization of incorrect methods or materials could result in a failure to restore vehicle's original structural integrity. Removal of some undamaged urethane bonded glass for reuse may not be possible due to damaging plastic locating studs and/or attached moldings. Some vehicle manufacturers recommend replacing glass that has been removed with new OEM glass. It is MOTOR's position that removal of the glass from the damaged part / panel for reuse is a process best reserved for the judgment of an estimator / appraiser following a thorough review of vehicle manufacturer guidelines.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **131. Suspension / Wheel Alignment**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/Suspension\\_WheelAlignment.htm](https://help.cccis.com/webhelp/motor/gte/Content/Suspension_WheelAlignment.htm)

### **Suspension / Wheel Alignment**

#### **SPECIAL NOTATION:**

Operations outlined in the following section are based on four wheel sensor alignment equipment. Alignment time includes normal wear diagnosis. It does not include collision related damage diagnosis.

#### **FRONT WHEEL ALIGNMENT**

Align front wheels

Center steering wheel

Check vehicle height & thrust angle

Check and adjust tire pressure

Check wheel size & run out

Remove vehicle from alignment equipment

Road test vehicle

Set up on alignment equipment & normal wear diagnosis

Shimming as required

Adjust brakes

Adjust vehicle height

Align rear wheels

Alignment shim cost

#### **REAR WHEEL ALIGNMENT**

Align rear wheels

Check and adjust tire pressure  
Check vehicle height & thrust angle  
Check wheel size & run out  
Remove vehicle from alignment equipment  
Road test vehicle  
Set up on alignment equipment & normal wear diagnosis  
Shimming as required  
Adjust vehicle height  
Adjust brakes  
Align front wheels  
Alignment shim cost

#### 4 WHEEL ALIGNMENT

Align front & rear wheels  
Center steering wheel  
Check vehicle height & thrust angle  
Check wheel size & run out  
Check and adjust tire pressure  
Remove vehicle from alignment equipment  
Road test vehicle  
Set up on alignment equipment & normal wear diagnosis  
Shimming as required  
Adjust brakes  
Adjust vehicle height  
Alignment shim cost

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **132. Tailgate - Passenger Cars / Wagons**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/TAILGATE.htm>

### **Tailgate - Passenger Cars / Wagons**

SPECIAL NOTATION:

Some replacement components may or may not be supplied with duplicated OEM caulk / seam sealer, this is not included in R&R time and requires an on-the-spot evaluation. Disconnect wiring at connector if accessible.

Disconnect at hinge & support cylinder.

Align to vehicle

Applique panel

Bonded glass installation

Electrical wiring (standard features)

Glass (moveable)

Finish panel

Finish plate

Finish molding

Finish trim

Inside release

Interior trim

Internal mechanisms

Lamps

Latch or striker attached to gate

License plate (if attached)

License pocket

Lock & cylinder assembly (if necessary)

Outside handle

Trim moldings

Trim panel R&I

Watershield

Weatherstrips (if attached to lift gate)

Window / run channel

Back glass wiper or washer

Bonded glass removal from damaged tailgate

Camera

Electrical wiring (if not detachable from main body harness)

Emblems & nameplates

Hinges

Lock cylinder recoding

Moldings (except as listed above)

Pinch sensors

Removal of shipping brace(s)

Spare tire carrier

Special interior trim

Spoiler, flare or louver

Stripe tape, decals or overlays

Things to Consider

Add If Required

Adhesive Type Moldings, Nameplates & Emblems

Damaged or Non-Reusable Moldings and Fasteners

Decals, Stripe Tape & Overlays

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

R & I (Remove & Reinstall)

Special Substrate Metals

Structural Glass

Refinish Starting Point

Basic Color Coat Application

Parts Supplied In Color

Refinish Clear Coat Application

Refinish Time Listings

Refinish Time Premise

Refinish Processes

Clear Coat Finishes (Base Coat / Clear Coat)

Color Blend (Adjacent Panels)

Deductions to Basic Refinish Times (Refinish Overlap)

Panel and/or Component Designation

Two-Tone Refinishing (Second Color Tone Application)

Underside of Hoods, Lids or Gates

Things to Know

Bagging (Cover Entire Vehicle Exterior)

Clear Coat Undamaged Panel

De-Nib & Polish

Edges Of New Parts (Edging)

Stone Chip Guard (Protective Material)

Three-Stage Color Blend (Adjacent Panels)

Three-Stage Finishes (Base / Mica / Clear Coat)

Wet / Dry Sand, Rub-Out & Buff

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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### **133. Three-Stage Color Blend (Adjacent Panels)**

Source:

[https://help.cccis.com/webhelp/motor/gte/Content/THREE\\_STAGE\\_COLOR\\_BLEND\\_Adjacent\\_Panels.htm](https://help.cccis.com/webhelp/motor/gte/Content/THREE_STAGE_COLOR_BLEND_Adjacent_Panels.htm)

### **Three-Stage Color Blend (Adjacent Panels)**

Estimated refinish times for color blending should defer to the judgment of an estimator or appraiser following an on-the-spot evaluation of the specific vehicle and refinish requirements in question.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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### **134. Trunk Lid**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/TRUNKLID.htm>

### **Trunk Lid**

SPECIAL NOTATION:

Some replacement components may or may not be supplied with duplicated OEM caulk / seam sealer. This is not included in R&R time and requires an on-the-spot evaluation.

Disconnect wiring at connector if accessible.

Disconnect at hinge.

Align to vehicle

Applique panel

Electrical wiring (standard features)

Finish panel

Finish plate

Finish molding

Finish trim

Inside release

Interior trim

Internal mechanisms

Lamps

Latch or striker attached to lid

License plate (if attached)

License pocket

Lock & cylinder assembly (if necessary)

Trim moldings

Weatherstrips (if attached to lid)

Camera

Electrical wiring (if not detachable from main body harness)

Emblems & nameplates

Hinge

Lock cylinder recoding

Moldings (except as defined in Included)

Release switch

Spoilers & flares

Stripe tape, decals or overlays

Things to Consider

Add If Required

Adhesive Type Moldings, Nameplates & Emblems

Decals, Stripe Tape & Overlays

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

Refinish Starting Point

Basic Color Coat Application

Parts Supplied In Color

Refinish Clear Coat Application

Refinish Time Listings

Refinish Time Premise

Refinish Processes

Clear Coat Finishes (Base Coat / Clear Coat)

Color Blend (Adjacent Panels)

Deductions to Basic Refinish Times (Refinish Overlap)

Panel and/or Component Designation

Underside of Hoods, Lids or Gates

Things to Know

Bagging (Cover Entire Vehicle Exterior)

Clear Coat Undamaged Panel

De-Nib & Polish

Edges Of New Parts (Edging)

Stone Chip Guard (Protective Material)

Three-Stage Color Blend (Adjacent Panels)

Three-Stage Finishes (Base / Mica / Clear Coat)

Wet / Dry Sand, Rub-Out & Buff

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **135. Tailgate - Outer Panel R&R**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/TailGate\\_OuterPanelR\\_R.htm](https://help.cccis.com/webhelp/motor/gte/Content/TailGate_OuterPanelR_R.htm)

### **Tailgate - Outer Panel R&R**

#### **SPECIAL NOTATION:**

Some cutting or welding processes may require the door glass to be removed prior to replacing the outer panel. This factor should be considered prior to finalizing any repair cost estimates. Disconnect wiring at connector if accessible.

Adhesive application if required (all types)

Align to vehicle

Applique panel

Caulk / seam sealer

Cutting & welding as necessary

Finish panel

Finish plate

Finish molding

Finish trim

Flutter material application if required

Grind, fill and smooth welded seams (up to 150 grit sandpaper)

Interior trim

Lamp

License plate (if attached)

License pocket

Lock cylinder assembly (if necessary)

Outside handle

Tailgate R&I

Trim panel R&I

Water shield

Weatherstrips (if attached)

Back glass wiper or washer

Bonded glass removal from damaged tailgate

Camera

Electrical wiring (if not detachable from main body harness)

Emblems & nameplates

Glass

Hinge

Moldings (except as defined in Included)

Reinforcement bar

Pinch sensors

Spare tire carrier

Spoiler, flare or louver

Straightening of frame

Stripe tape, decals or overlays

Things to Consider

Add If Required

Adhesive Type Moldings, Nameplates & Emblems

Damaged or Non-Reusable Moldings and Fasteners

Decals, Stripe Tape & Overlays

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

R & I (Remove & Reinstall)

Special Substrate Metals

Structural Glass

Refinish Starting Point

Basic Color Coat Application

Parts Supplied In Color  
Refinish Clear Coat Application  
Refinish Time Listings  
Refinish Time Premise  
Refinish Processes  
Clear Coat Finishes (Base Coat / Clear Coat)  
Color Blend (Adjacent Panels)  
Deductions to Basic Refinish Times (Refinish Overlap)  
Panel and/or Component Designation  
Two-Tone Refinishing (Second Color Tone Application)  
Underside of Hoods, Lids or Gates  
Things to Know  
Bagging (Cover Entire Vehicle Exterior)  
Clear Coat Undamaged Panel  
De-Nib & Polish  
Edges Of New Parts (Edging)  
Stone Chip Guard (Protective Material)  
Three-Stage Color Blend (Adjacent Panels)  
Three-Stage Finishes (Base / Mica / Clear Coat)  
Wet / Dry Sand, Rub-Out & Buff

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## **136. Three-Stage Finishes (Base / Mica / Clear Coat)**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/ThreeStageFinishes\\_Base\\_Mica\\_ClearCoat\\_.htm](https://help.cccis.com/webhelp/motor/gte/Content/ThreeStageFinishes_Base_Mica_ClearCoat_.htm)

### **Three-Stage Finishes (Base / Mica / Clear Coat)**

#### **SPECIAL NOTATION:**

The following items or operations were not considered during the development of any published basic refinish operation times. If any of these items or operations are required, they should be considered by the estimator. Calculations should be made after deduction for overlap, and additions for underside and edges if required. (if three-stage finish from factory).

- First major panel;
- Add 70% to refinish time

- Each additional panel;

– Add 40% to refinish time

Back tape opening (handle, lock cylinder, mirror)

Clear coat application

Mask / close gap between adjacent panels up to foam tape (over spray)

Mask glass opening

Mask / protect grille radiator opening (over spray)

Retrieve accurate color information, including paint chip

Tinted clear as required by paint manufactures color formula

Any component clear coated as a separate procedure

Any operation previously excluded in "Refinish Time Premise" and/or "Basic Color Coat Application" groups

Flash/wait time between material applications

Material costs

Test spray-out panel

Tint clear coat to achieve color match

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **137. Two-Tone Refinishing (Second Color Tone Application)**

Source:

[https://help.cccis.com/webhelp/motor/gte/Content/TwoToneRefinishing\\_SecondColorToneApplication\\_.htm](https://help.cccis.com/webhelp/motor/gte/Content/TwoToneRefinishing_SecondColorToneApplication_.htm)

### **Two-Tone Refinishing (Second Color Tone Application)**

SPECIAL NOTATION:

The following items or operations were not considered during the development of any published basic refinish operation times. If any of these items or operations are required, they should be considered by the estimator. The following formula may be superseded by information published in specific parts text. Calculations should be made after deductions for overlap, and additions for edges if required.

- First panel;
- Add 40% to refinish time
- Each additional panel;
- Add 30% to refinish time
- Clean sprayer
- Color coat application

- Load sprayer
- Masking for second color
- Mix paint (color with necessary solvents)
- Remove masking
- Retrieve accurate color information, including paint chip
- Backside refinishing
- Blending into adjacent panels
- Clear coat
- Color tinting
- Mixing a different edge color
- Paint or material costs
- Prime & block (high build/primer-filler)
- Test spray-out panel
- Tinting Primer-Sealer

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **138. Underhood Diagrams**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/UnderhoodDiagrams.htm>

### **Underhood Diagrams**

We include underhood diagrams when available. All units of measurements are millimeters. Please refer to MOTOR "Collision Repair Data" manual or MOTOR E-Frame Data on CD for complete vehicle dimension specifications. All underhood illustrations used herein are copyrighted by Hein-Werner Corporation, Milwaukee Wisconsin and used with the permission of the copyright owner.

Underside of Hoods, Lids or Gates

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## **139. Underside of Hoods, Lids or Gates**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/UndersideOfHoods\\_LidsOrGates.htm](https://help.cccis.com/webhelp/motor/gte/Content/UndersideOfHoods_LidsOrGates.htm)

## **Underside of Hoods, Lids or Gates**

### **SPECIAL NOTATION:**

The following items or operations were not considered during the development of any published basic refinish operation times. If any of these items or operations are required, they should be considered by the estimator. An additional paint mix is required if the underside color is a different color than the exterior body color. Clear coat (gloss or matte) will be required for base color coat applications.

Refer to specific vehicle text for estimated time allowance

Use full refinishing time without deduction for overlap

Clear coat

Color tinting

Mixing a different underside color

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **140. Uniside with Quarter Panel**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/UnisideWithQuarterPanel.htm>

### **Uniside with Quarter Panel**

#### **SPECIAL NOTATION:**

When replacing welded door hinges, add one half of the Door Hinge R&R time to Uniside R&R time. Included and Does Not Include: items also apply to sectioning below roof (lower uniside) and sectioning roof rail (upper uniside). When sectioning other areas of uniside, follow GTE page for portion of uniside being sectioned.

Adhesive application if required (all types)

Back glass & moldings

Carpet & insulation turn back

Caulk / seam sealer

Center pillar trim (4 door models)

Cutting and welding as necessary

Door striker

Front seat R&I

Fuel Door

Grind, fill & smooth welded seams (up to 150 grit sandpaper)

High-mounted stop lamp (if necessary)

Hinge pillar glass and moldings  
Hinge pillar trim  
Jamb switches  
Package tray trim  
Quarter panel trim  
Quarter glass & moldings  
Rear bumper R&I  
Rear lamp  
Rear seat R&I  
Rivet removal and installation if required  
Roof drip rail moldings  
Seat belt anchor  
Sill plates  
Trunk compartment trim  
Weatherstrips  
Wheelhouse liner (plastic)  
Airbag components  
Body side moldings  
Carpet & insulation R&I  
Control cables  
Doors  
Door hinge  
Electrical wiring  
Emblems & nameplates  
Fender  
Foam  
Fluid lines  
Headliner  
Instrument panel  
Road wheel  
Roof cover  
Stripe tape, decals or overlay  
Windshield glass and moldings  
Wiring

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **141. Uniside / Aperture Panel Without Quarter Panel**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/UnisideWithoutQuarterPanel.htm>

### **Uniside / Aperture Panel Without Quarter Panel**

#### **SPECIAL NOTATION:**

When replacing welded door hinges, add one half of the Door Hinge R&R time to Uniside R&R time. Included and Does Not Include: items also apply to sectioning below roof (lower uniside) and sectioning roof rail (upper uniside). When sectioning other areas of uniside, follow GTE page for portion of uniside being sectioned.

#### **Included:**

Adhesive application if required (all types)

Carpet & insulation turn back

Caulk / seam sealer

Center pillar trim (4 door models)

Cutting and welding as necessary

Door striker

Front seat R&I

Grind, fill & smooth welded seams (up to 150 grit sandpaper)

Hinge pillar glass and moldings

Jamb switches

Package tray trim

Quarter panel trim

Rear seat R&I

Roof drip rail moldings

Rivet removal and installation if required

Seat belt anchor

Sill plates

Weatherstrips

Airbag components

Back glass and moldings

Carpet & insulation R&I

Control cables

Doors

Door hinge  
Foam  
Electrical wiring  
Emblems & nameplates  
Fender  
Fluid lines  
Headliner  
Instrument panel  
Quarter glass and moldings  
Road wheel  
Roof cover  
Stripe tape, decals or overlay  
Windshield glass and moldings

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **142. Unprimed Flexible Component Preparation**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/UnprimedBumperPrep.htm>

### **Unprimed Flexible Component Preparation**

- 25% of the component's base refinish time
- Maximum time allocation: 1.0 hours

Removal of mold-release agents as outlined by manufacturer

Masking (if required)

Application of adhesion promoter

Correction of pre-existent surface imperfections

Material Costs

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **143. Valance Panel**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/VALANCE\\_PANEL.htm](https://help.cccis.com/webhelp/motor/gte/Content/VALANCE_PANEL.htm)

## **Valance Panel**

Brackets / Braces

Lamps

Grilles

Air dam

Bumper assembly

Emblems & nameplates

Moldings

Spoilers & flares

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **144. Van / SUV - Side & Corner Panels**

Source: [https://help.cccis.com/webhelp/motor/gte/Content/Van\\_Truck\\_Side\\_CornerPanels.htm](https://help.cccis.com/webhelp/motor/gte/Content/Van_Truck_Side_CornerPanels.htm)

### **Van / SUV - Side & Corner Panels**

#### **SPECIAL NOTATION:**

Side and corner panel R&R times are generally based upon replacing at the factory seams. On some vehicles, it is not practical to replace at the factory roof seam. If a sectioning operation is not listed, then the R&R time on the parts line represents the time to splice at the best area. When a sectioning operation time is listed, it represents the time to splice at the most efficient location, usually within the window openings. Both Full Panel R&R operations and Partial Panel Sectioning operations involve the same basic "INCLUDED" and "NOT INCLUDED" procedure steps.

Adhesive application if required (all types)

Bolt on extensions & fillers

Bumper assembly R&I

Carpet & insulation turn back

Caulk/seam sealer

Cutting & welding as necessary

Detachable roof assemblies

Door striker

Fuel door

Grind, fill & smooth welded seams (up to 150 grit sandpaper)

Interior trim

Lamp

Rivet removal and installation if required

Side glass & moldings

Sill plates

Sliding door tracks (bolted)

Weatherstrips & seals

Wheelhouse liner

ABS sensor

Airbag components

Body side moldings

Brackets, braces & supports transfer

Carpet & insulation R&I

Emblems & nameplates

Exhaust system

Floor

Foam

Fuel tank

Headliner

Mud guard

Second row seats

Sliding door tracks (welded)

Spare tire carrier

Stripe tape, decals or overlays

SUV rear hinged door

Tailgate / liftgate

Third row seats

Van sliding door

Wheelhouse

Window (aftermarket)

Things to Consider

Acoustical and Structural Foam Fillers

Add If Required

Adhesive Type Moldings, Nameplates & Emblems

Bolted On Parts

Component Classification

Damaged or Non-Reusable Moldings and Fasteners  
Decals, Stripe Tape & Overlays  
Drilling Operations  
Footnotes  
Included and/or Not Included Labor Operations  
Labor Time Listings  
Labor Time Premise  
Panel Bonding (Metal Adhesives)  
R & I (Remove & Reinstall)  
R & R (Remove & Replace)  
Sectioning  
Special Substrate Metals  
Structural Glass  
Refinish Starting Point  
Basic Color Coat Application  
Parts Supplied In Color  
Refinish Clear Coat Application  
Refinish Time Listings  
Refinish Time Premise  
Refinish Processes  
Clear Coat Finishes (Base Coat / Clear Coat)  
Color Blend (Adjacent Panels)  
Deductions to Basic Refinish Times (Refinish Overlap)  
Panel and/or Component Designation  
Two-Tone Refinishing (Second Color Tone Application)  
Things to Know  
Bagging (Cover Entire Vehicle Exterior)  
Bolt Or Clip-On Moldings  
Clear Coat Undamaged Panel  
De-Nib & Polish  
Edges Of New Parts (Edging)  
Panel Bonding (Metal Adhesives)  
Stone Chip Guard (Protective Material)  
Three-Stage Color Blend (Adjacent Panels)  
Three-Stage Finishes (Base / Mica / Clear Coat)  
Weld Zone / Adjacent Panel  
Wet / Dry Sand, Rub-Out & Buff

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **145. WELDER SET-UP AND WELDING PREPARATION**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/WELDER SET-UP AND WELDING PREPARATION.htm>

### **WELDER SET-UP AND WELDING PREPARATION**

Due to the different types of welding equipment used in the collision repair industry, labor times for welded replaced parts do not include equipment manufacturer procedural steps for welder setup and/or welding tests and preparation. Each welding machine manufacturer may have its own unique configurations and setup processes. Additionally, there may be vehicle-specific variables that may increase or decrease the amount of welding machine set-up time and pre-weld preparation. MOTOR suggests using an on-the-spot evaluation to determine an appropriate set-up and preparation time.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## **146. Weld Zone / Adjacent Panel**

Source: <https://help.cccis.com/webhelp/motor/gte/Content/WeldAreaBlendZones.htm>

### **Weld Zone / Adjacent Panel**

#### **SPECIAL NOTATION:**

Suggested refinish operation times do not include additional time for repair of damage to adjacent panels resulting from normal cutting, welding and grinding procedures. The amount of damage can vary considerably depending upon process and technique used by the servicing technician, and therefore is impractical to anticipate in this publication. MOTOR recommends these factors be considered before finalizing any repair cost estimate. Typical areas to be considered are illustrated below.

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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## 147. Wet / Dry Sand, Rub-Out & Buff

Source: [https://help.cccis.com/webhelp/motor/gte/Content/WetSand\\_Rub\\_OutAnd\\_OrBuff.htm](https://help.cccis.com/webhelp/motor/gte/Content/WetSand_Rub_OutAnd_OrBuff.htm)

### Wet / Dry Sand, Rub-Out & Buff

#### SPECIAL NOTATION:

Refinished panels may or may not require a varying amount of wet sanding, compound rub-out, or buffing operations in order to match original finish texture. The clear coat contains ultraviolet screeners and reducing the clear coat thickness (mils) may result in early paint failure. Follow manufacturer's recommendations when performing this type of repairs. Calculations should be based upon the outer surface only and should not include additions for clear coat, underside, inside or edges. Base refinish time does not include deduction for refinish overlap. In the event that this type of operation will be performed, MOTOR suggests the following formula be considered:

Refinished panels may or may not require a varying amount of wet sanding, compound rub-out, or buffing operations. In the event this type of operation will be performed, MOTOR suggest the following formula be considered.

- Each panel requiring wet sand, rub-out and/or buff (refinish or blend)
  - Add 30% of full base refinish time

Compound, buff and/or polish as required

Panel outer surface only

Wet sand full panel as required

Acid rain damage

Overspray removal

Removal of residual material from recessed edges and jambs if required

Wash, clean, wax or detail entire vehicle prior to delivery if required

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## 148. Windshield Glass

Source: <https://help.cccis.com/webhelp/motor/gte/Content/WindshieldGlass.htm>

### Windshield Glass

#### SPECIAL NOTATION:

Glass, moldings, mounting studs or mounting flange may be damaged or broken during normal service procedures on some applications. MOTOR advises that these factors be considered before performing glass operations in the event unavoidable damage occurs. Time for the removal of old adhesive, clean and

preparation of sealing surfaces (vehicle and/or glass) should be estimated after an on the spot evaluation.

Cowl vent panel / screen (if overlapping glass)

Rear view mirror / support (attached to glass)

Reveal & interior moldings / trim

Rubber channel R&I (if installed)

Test for leaks

Urethane adhesive application

Wiper arms

Camera

Clean up of broken glass

Cost of installation material or kit

Rain sensors

Refinish glass pinchweld (epoxy primer)

Things to Consider

Adhesive Type Moldings, Nameplates & Emblems

Decals, Stripe Tape & Overlays

Footnotes

Included and/or Not Included Labor Operations

Labor Time Listings

Labor Time Premise

NAGS - Glass Prices

R & I (Remove & Reinstall)

Structural Glass

Footnotes found in a chapter contain vehicle-specific information. The content of footnotes is in addition to, and takes precedence over, information in the Guide to Estimating pages for the operation indicated.

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